When Better Teachers Aren't Enough:

An Experimental Evaluation of Teacher Training Programs in El Salvador

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Despite billions spent annually on teacher training, rigorous evidence on standalone in-service programs remains scarce, as most evaluated programs bundle training with curriculum or material reforms. We address this gap through a large-scale randomized controlled trial with 338 schools and over 6,000 students in El Salvador. Teachers are randomly assigned to either a control group or one of three training programs focusing on (i) content knowledge, (ii) pedagogical knowledge, or (iii) a combination of both inputs. We find lasting effects on teacher content and pedagogical knowledge of up to 0.3σ and 0.57σ respectively one year after program end. Yet, this only changes teachers' classroom practices in the short-run and does not translate into significant student learnings. The data most closely aligns with a setting where teachers face a dual challenge: sustaining new instructional practices in a rigid environment while navigating the significant learning gaps present among students in later grades.

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1 Introduction

Quality education remains elusive in many developing countries, even as school enrollment has expanded substantially in recent decades. International assessments consistently show that the rate at which education systems convert schooling into human capital is low. Recent estimates suggest that around 70% of children in low- and middle-income countries, and nearly 80% in Latin America and the Caribbean, are unable to read and understand a simple text by age 10 (World Bank, 2022). This has prompted the World Bank to dedicate its 2018 World Development Report to what was declared a global "learning crisis" (World Bank, 2018). Evidence from Africa, Asia, and Latin America indicates that shortcomings in teachers' pedagogical and content knowledge are a major barrier to more effective schooling systems, implying that the learning crisis is rooted in a deeper teaching crisis (Bold et al., 2017a; Bruns and Luque, 2014; Molina et al., 2018). Without joint efforts, this situation is likely to reproduce itself: Many of today's poorly qualified teachers will continue teaching for years to come and consequently shape tomorrow's teachers.

Improving teacher quality is therefore a stated policy priority across developing countries. Governments devote billions of dollars each year to professional development (Loyalka et al., 2019), and the latest available UNESCO data show that roughly four in five teachers report receiving in-service training within the previous year across income groups (UNESCO Institute for Statistics, 2026). Yet the existing evidence base is strikingly misaligned with policy practice. Governments largely train teachers within existing instructional systems, whereas the literature disproportionately evaluates interventions that alter those systems, simultaneously changing what teachers teach, the materials they use, or the technologies and support available to them. In practice, only about one third of large, at-scale government programs provide scripted lessons or instructional materials (within the cross-country sample studied by Popova et al., 2022), while nearly 70% of estimates in the causal evaluation literature bundle training with new curricula or materials and another 21% study other specialized interventions (Figure A1). As a result, despite a sizable literature on teacher training, evidence on the conventional in-service models that dominate public-sector provision remains surprisingly thin.

We address this gap through a large-scale randomized controlled trial (RCT) across 338 schools in eastern El Salvador, a setting characterized by low levels of both teacher pedagogical and content knowledge. We examine whether well-implemented, standalone in-service training can improve student learning, and whether impacts

differ by an emphasis on pedagogical versus content knowledge. We randomly assign 338 upper-primary math teachers to either a control group or one of three training programs focusing on (i) content knowledge, (ii) pedagogical knowledge, or (iii) a combination of both inputs. Because each treatment arm follows the same six-month, seven-block structure, the combined arm blends content and pedagogy within a fixed training dosage. All training programs share a common basic framework combining face-to-face workshops, coaching elements, and self-study modules. The program was developed in collaboration with educational experts from the Swiss University of Teacher Education Fribourg and Consciente, a Salvadorian NGO specializing in evidence-based schooling projects. It was approved and supported by the Salvadorian Ministry of Education.

To evaluate the effects of the interventions and examine the mechanisms through which they operate, we assemble rich data on teacher knowledge, instructional behavior, teacher beliefs and attitudes and student achievement over two school years. Outcomes are measured at baseline, immediately after program completion (endline), and one year later (follow-up). Teacher content knowledge is assessed using math tests that span the full primary math curriculum, with particular emphasis on higher-grade material. Pedagogical knowledge is measured at follow up using a paper based assessment with two complementary tasks: lesson plan design for a specified topic and classroom context, and responses to a classroom management scenario. Instructional practices are measured at endline and follow-up through classroom observations based on a modified version of the Stallings instrument, which records instructional time allocation and includes additional low-inference pedagogical items (Stallings, 1973). We complement these measures with teacher surveys at endline and follow-up covering lesson preparation, perceived teaching competence, motivation, and perceived effects on student learning, as well as semi-structured interviews with treated teachers to provide qualitative evidence on teachers' experiences with the program. Finally, student achievement is captured using curriculum-aligned math tests administered to one fourth-grade and one fifth-grade classroom per teacher.

Baseline descriptive evidence reveals severe shortfalls in math knowledge among both students and teachers. At the beginning of fourth and fifth grade, students answer 72% of first-grade items correctly, and accuracy drops by more than half—to 35%—for second-grade items and further to 26% and 12% for third- and fourth-grade items respectively. Teachers likewise demonstrate incomplete mastery of the primary curriculum, correctly answering just 61 percent of grade two to six items on average, in line with existing evidence from El Salvador (Brunetti et al., 2024). To benchmark

these outcomes internationally, we map student and teacher performance onto the standardized Trends in International Mathematics and Science Study (TIMSS) scale using Item Response Theory (IRT). Salvadorian students rank near the bottom of the global distribution, second only to the lowest-performing country in the TIMSS sample. Notably, the average Salvadorian teacher barely outperforms the average fourth-grade student in Singapore, the top-performing system in TIMSS, underscoring the depth of content knowledge gaps in the education system.

To benchmark ex ante expectations, we elicited predictions from education researchers and practitioners on the social science prediction platform. Based on a detailed description of the intervention and its theory of change, respondents predicted sizable improvements in teacher knowledge, classroom practices, as well as student learning across all treatment arms. Mean predicted effects for student achievement were around 0.15σs, marginally smaller than expected teacher gains ($\sim$0.16–0.17σ) but still substantial. Only four out of 38 respondents anticipated student effects below 0.05σs, and among them, just one predicted negligible student gains while expecting large improvements for teachers. These priors suggest that most informed observers expected teacher trainings to improve student learning, and that the few skeptical views were driven by doubts about teacher learning rather than a failure of knowledge transmission.

In line with expectations, we find substantial effects on teachers across all three programs. The math program increases content knowledge by 0.14σ at endline and 0.29σ one year later the combined program yields smaller but similarly growing gains (0.11σ and 0.15σ); the pedagogy-only program, consistent with its design, produces no measurable improvement in content knowledge. The long-run gain of 0.29σ corresponds to roughly 23 points on the TIMSS scale, comparable to the achievement gap between Salvadorian fourth- and fifth-graders or the difference in fourth-grade country averages between Chile and France. Across all arms, pedagogical knowledge, as measured by the paper based assessment, improves substantially at follow-up, ranging from 0.47σ to 0.57σ. These gains, however, do not translate into student learning. We detect no gains in student achievement immediately after implementation or one year later; estimates are consistently small across treatments, waves, and outcome specifications, with even the largest positive estimate allowing us to rule out effects above approximately 0.16σ at the 95% confidence level.

Drawing on our complementary classroom observations, teacher survey data and semi-structured interviews, we examine four key steps linking teacher trainings to student learning: (i) successful implementation and high compliance, (ii) meaning-

ful improvements in teacher knowledge, (iii) translation of these gains into classroom practice, and (iv) students' ability to absorb and retain instruction. First, the intervention achieved high implementation fidelity, with over 80% of teachers participating and semi-structured interviews confirming high engagement and strong acceptance of the program. Second, the training substantially improved teachers' content and pedagogical knowledge. Third, these gains translated into broad improvements in classroom instruction at endline, including more interactive classroom activities (+12–18 percentage points; +140–200% relative to the control mean), greater use of interactive teaching materials (+4–5 percentage points; +110–130%), lower classroom distraction (-9 percentage points; -20%), and higher scores on a composite index of observed instructional practices ($+0.32\sigma$). However, these changes largely dissipated by the one-year follow-up, and teacher surveys and semi-structured interviews suggest that the withdrawal of ongoing pedagogical accompaniment may have contributed to this fade-out. Fourth, and perhaps most critically, a sharp misalignment between the prescribed curriculum and students' actual learning levels points to a major bottleneck to student progress. We find heterogeneous treatment effects by baseline achievement: students who are at most one year below grade-level proficiency at baseline gain approximately 0.3σ in both the mathematics and pedagogy treatment arms, while effects for students further behind are statistically indistinguishable from zero.

This study contributes to the literature in four ways. First, we clarify an important but often overlooked distinction in the teacher-training literature and provide new evidence on conventional in-service training. Standard in-service training seeks to improve teachers within the existing instructional system, whereas structured-pedagogy reforms simultaneously alter not only how teachers teach but also what they teach. Many of the most influential positive results attributed to teacher training come from this latter class of interventions, which typically combine training with revised curricula, structured materials, and ongoing coaching (see, e.g., Kerwin and Thornton, 2021; Buhl-Wiggers et al., 2023; Cilliers et al., 2020; Bruns et al., 2018; Bassi et al., 2020; Albornoz et al., 2020; Piper et al., 2018; Jukes et al., 2017). Angrist et al. (2025) are among the first to make this distinction explicit by separating structured-pedagogy reforms from other interventions involving teacher training. The remaining literature is itself heterogeneous, encompassing technology- or coaching-centered interventions (e.g., Castro et al., 2025; Bruns et al., 2018; Jacob, 2017; Barrera-Osorio and Linden, 2009), training on targeted topics such as HIV, financial literacy, or entrepreneurship (e.g., Duflo et al., 2015; Blimpo and Pugatch, 2021; Compen et al., 2023, 2021), and programs built around specific pedagogical approaches such as Teaching at the

Right Level, learning-to-learn, or student-centered instruction (e.g., Banerjee et al., 2016; Ashraf et al., 2020; Jakob et al., 2023). Our review of recent evidence syntheses (Popova et al., 2022; Angrist et al., 2025), supplemented by a keyword search of the Social Science Registry, shows how small the evidence base on conventional training is: only 9% of eligible experimental estimates—seven estimates from four papers—evaluate conventional stand-alone in-service training, and only five estimates from two papers concern core academic subjects like mathematics, science or language (FigureA1). We add a large-scale randomized evaluation of a well-implemented mathematics training program that strengthens teachers' knowledge and practice without changing the curriculum or introducing new student-facing instructional materials.

Second, we show that effective implementation and substantial teacher learning may not be sufficient for teacher training to improve student achievement. Existing evaluations of core-subject in-service training find null effects in settings with limited take-up or weak implementation (Loyalka et al., 2019; Schaffner et al., 2025), leaving open whether conventional training can work when these constraints are relaxed. Our NGO-led program provides such a test: participation is high, teacher content and pedagogical knowledge improve substantially and persistently, and classroom practice improves in the short run. Yet average student achievement remains unchanged. Rich intermediate outcomes allow us to trace this disconnect further downstream. Instructional changes fade once support is withdrawn, while learning gains are concentrated among students sufficiently close to grade-level content to benefit from improved instruction.

Third, we contribute to understanding what teacher training should target. Teachers in many developing-country settings exhibit substantial deficits in subject-matter knowledge (Bold et al., 2017b; Sinha et al., 2016; Brunetti et al., 2024), and existing evidence suggests that these deficits are consequential for student learning, particularly in mathematics (Metzler and Woessmann, 2012; Bold et al., 2019). Yet most evaluated professional-development programs emphasize pedagogy rather than subject mastery, leaving limited causal evidence on how to improve teachers' content knowledge (Bold et al., 2017b). We address this gap by holding total training duration fixed while varying its focus across mathematics content, pedagogy, and a combination of the two. This design identifies the returns to specialization in content versus pedagogy and tests whether a blended program can exploit complementarities between them. While existing work shows that educational inputs can be complementary (Mbiti et al., 2019), whether content and pedagogical training reinforce one another remains largely unknown. Our design therefore provides a direct policy-relevant

test of whether such complementarities are strong enough to offset the lower intensity devoted to each component in a combined program.

Fourth, we contribute to recent efforts to improve the interpretability and comparability of evidence across educational interventions (Angrist et al., 2025). Building on Patel and Sandefur (2020), we benchmark teacher and student achievement to the international TIMSS scale, allowing both baseline learning levels and treatment effects to be interpreted relative to internationally recognized learning standards rather than study-specific standardized units. Using an externally anchored benchmark facilitates meaningful comparisons across interventions, countries, and populations, and complements recent efforts to document learning levels on a globally comparable scale (Angrist et al., 2021; Patel and Sandefur, 2020).

2 Research Design

2.1 Context and Interventions

Our study is set in El Salvador, a lower-middle-income country in Central America. It was conducted in its oriental region which comprises 4 out of the 14 existing departments of the country: Morazán, Usulután, San Miguel and La Unión (see Figure 1). El Salvador's education system mirrors that of many developing countries: a substantial share of students complete primary school without basic literacy skills, and learning outcomes fall short of minimum proficiency standards (Altinok et al., 2018; Coccia et al., 2026). In fact, recent World Bank data shows that the quality of primary education in El Salvador falls below the average of lower middle-income countries (Angrist et al., 2021).

A prominent feature of El Salvador's education system is the low level of teachers' content knowledge and pedagogical skills. Teachers lack a basic mathematical understanding of the very concepts they are required to teach. Correspondingly, a recent study by Brunetti et al. (2024) finds that the average, representative teacher in Morazán, one of the study departments, answers more than half of grade two to six questions incorrectly. The study concludes, that a mere 14 percent of teachers are actually equipped with the necessary content knowledge to effectively teach mathematics at the primary school level.[2] Despite having completed 13 to 17 years of formal education, the majority of primary school teachers are therefore "confronted with the daunting task of teaching what they don't know" (Brunetti et al., 2024, p. 209).

[2]Based on the minimum proficiency threshold proposed by (Bold et al., 2017a).

At the same time, teachers' pedagogical knowledge appears limited. Practical pedagogy training does not form part of the official curriculum of El Salvador's national teacher training institutions in El Salvador. Consequences are evident: The traditional chalk-and-talk teaching style, centered almost exclusively on the blackboard and the teacher's exposition, continues to dominate Salvadorian classrooms. Consistent with this, classroom observations from the control group show that when teachers use instructional materials, 95% of the time they rely on conventional resources such as textbooks, notebooks, and the blackboard. Accordingly, teachers tend to be more focused on adhering to and covering the school curriculum as determined by the government textbook, rather than on the actual learning progress of their students.

(a) Sample departments (b) School locations

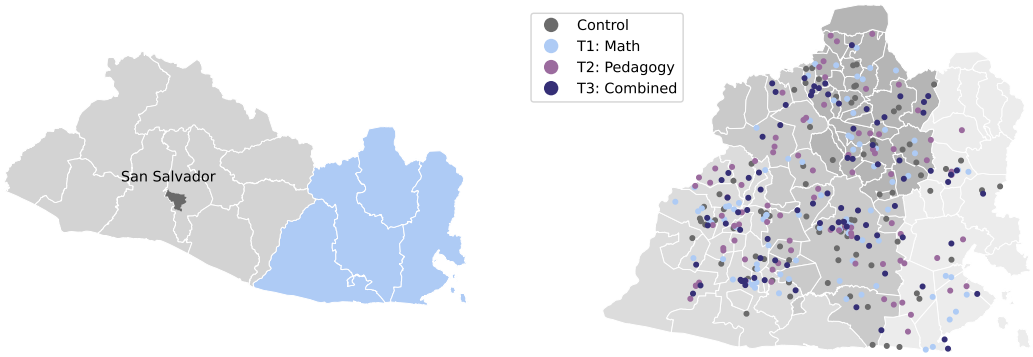


Figure 1: Study area

In this context, we partnered with the local NGO Consciente and experts from the Swiss University of Teacher Education Fribourg with the aim of strengthening teacher skills and enhancing student learning outcomes in public primary schools. We implemented a large-scale randomized control trial that compares three comprehensive teacher training programs, each targeting a distinct area: (i) mathematical content knowledge (ii) pedagogical skills and (iii) a combined approach.

Each intervention was delivered over a six-month period. The training structure was divided into seven three-week blocks that include a mix of in-person workshops, self-study, and personalized coaching (see Figure 2). Each block began with a full-day workshop (approximately 4.5 hours) held on a Saturday. Over the remainder of the three-week block, teachers completed approximately one hour of short instructional videos, around eight hours of guided self-study, and roughly one hour of individualized

online coaching with their assigned facilitator. The program and program materials were developed in a joint effort of Swiss and Salvadorian education experts and approved as well as supported by the Salvadorian Ministry of Education. The content of the training blocks for mathematics (i) was designed to prioritize the most relevant aspects of teaching upper primary grades while ensuring a foundational review of key mathematical principles. It covered topics including whole numbers, basic operations, fractions, decimals, geometry, and statistics.For pedagogy (ii), it covered essential didactic skills related to lesson planning, classroom management, student motivation, assessment, and differentiated instruction aimed at responding to diverse student needs and learning levels. The combined approach (iii) was streamlined to focus on the most impactful components of both the math and pedagogical interventions (a detailed overview of the covered topics can be found in Figure A2).[3]

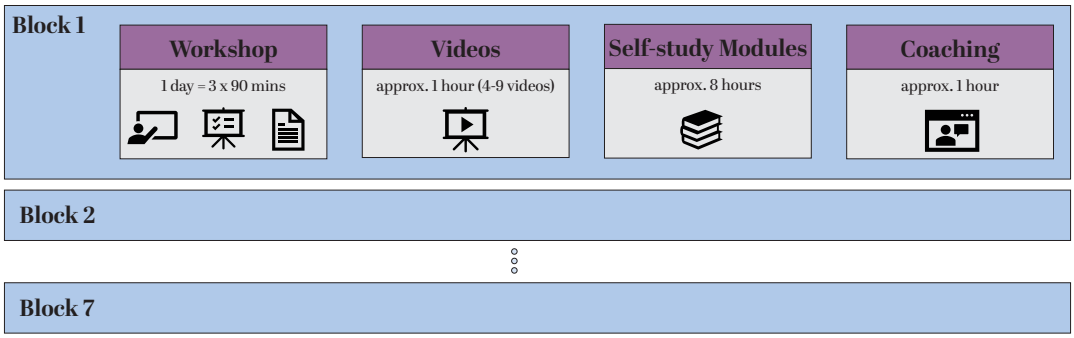


Figure 2: Program structure

Notes: Each training block has a duration of three weeks. In-person workshops are thus three weeks apart.

The trainings were explicitly designed to satisfy rigorous pedagogical standards and featured substantial interactive elements, including clear learning objectives, structured practice with feedback and collaborative problem-solving activities. To facilitate transfer to teachers' day-to-day instruction, all trainings followed the same instructional sequence used in El Salvador's nationally mandated ESMATE mathematics textbooks: introduction, content, practice, summary. The mathematics training followed the same design logic as the other two trainings, thereby enabling teachers not only to strengthen their content knowledge but also to benefit from more effective pedagogical strategies for conveying key concepts. Following the evidence in Popova et al. (2022) on most impactful training practices, the program incorporated

[3]Full materials available here: https://mdid.consciente.ong

structured lesson enactment as well as extensive in-person sessions. Additionally, in line with the same evidence, completion of the training was tied to career incentives through Ministry of Education certification. Participation was voluntary and teachers received no financial compensation.

The teacher trainings were conducted by ten teaching professionals employed by the partnering NGO Consciente. Prior to delivering the training, these instructors underwent an extensive training phase lasting several weeks, during which they were thoroughly acquainted with the training content and materials.[4] The training for the NGO-hired instructors was directly conducted by the Swiss-Salvadorian team that developed the teacher training program, ensuring that the implementation in the field reflected the intentions and vision of the training developers. Each instructor was assigned two teacher groups from the same intervention, resulting in group sizes of 10 to 15 participating teachers.

2.2 Sampling and Randomization

The target population of our study comprises all teachers who instructed math to at least one class in grades four or five in the first year of the study. Our teacher sample is based on voluntary registration for the program. To determine our sample, we first conducted school visits in all departments to inform school heads and teachers about the trainings and the scientific study. Teachers either registered voluntarily with the representative during the school visit or later via phone. In a second step, we invited all enlisted teachers to the baseline teacher assessment, where they provided written informed consent to participate in the study.

Based on the list of participants at the teacher assessment, we selected 338 teachers to participate in the study. Following Bloom et al. (2007), assuming 80% power and a 5% significance level, we obtain a minimum detectable effect (MDE) of 0.125.[5]

To avoid distortion of treatment effect estimates via spillovers (e.g., Miguel and Kremer, 2004), teachers were drawn from 338 different schools. Thus, we randomly selected only one teacher if two teachers from the same school enrolled. The resulting sample is not representative of the teacher population of our study area but mirrors standard teacher training arrangements in developing countries, where participation

[4]Beyond an initial two to three week review of the training materials, instructors had one to two weeks of preparation before each workshop with Consciente.

[5]The calculation assumes 84 teachers in both the treatment and control groups, an average of 18 students per teacher, an intra-class correlation of 0.20, a between-school R^2 of 0.68, and a within-school R^2 of 0.63. The latter parameters are calibrated using data from a prior study of teachers and students in Morazán (Büchel et al., 2022).

is often optional or weakly enforced.

Totaling 6,010 students at baseline, the student sample consists of all fourth and fifth grade students that receive math classes from the teachers in our sample. About half of the teachers only taught one of the target grades, which is why in those cases only one grade entered our sample. For the remainder, one class each from grades four and five participated in the study.[6]

We randomly assigned teachers to either a control group or one of three experimental groups receiving: (i) math training (ii) pedagogical training or (iii) the combined approach. The random assignment was stratified by the four study departments and terciles of baseline teacher performance. Overall, this yielded twelve strata containing 28 schools on average.[7] This randomization procedure led to assignment to training of (i) 85 teachers in math, (ii) 85 teachers in pedagogy, (iii) 84 teachers in the combined approach, and finally (iv) 84 teachers receiving no training in the control group.

2.3 Data

In all experimental groups, we conduct three measurement waves: a baseline measurement before the intervention, an endline measurement toward the end of the intervention, and a follow-up measurement one year after the intervention. This allows us to study both the immediate impact of the three interventions and whether potential effects are sustained one year after the program. Figure 3 shows the timeline of the project. The six month intervention took place in the school year 2022. The follow-up measurement was conducted at the end of 2023.

To track learning outcomes and explore potential program mechanisms, we collected an extensive set of data across teachers and students composed of: (i) teacher math and pedagogy tests, (ii) teacher classroom observation data, (iii) teacher surveys, (iii) program implementation data monitoring teachers' compliance with the program and gathering feedback on the programs, (iv) student math tests and finally (v) a short student survey.

2.3.1 Teacher Data

As one of the two main outcomes of our study, we administered both math and pedagogy tests to evaluate teachers' knowledge of the two subjects over time. Math

[6]In case a teacher instructed multiple classes on the same grade level, students of stream A, e.g. students of class 4A rather than class 4B, participated. However, this was very rarely the case.

[7]Strata leftovers were randomly assigned to treatment first within departments and finally, if still remaining as a leftover, on the national level.

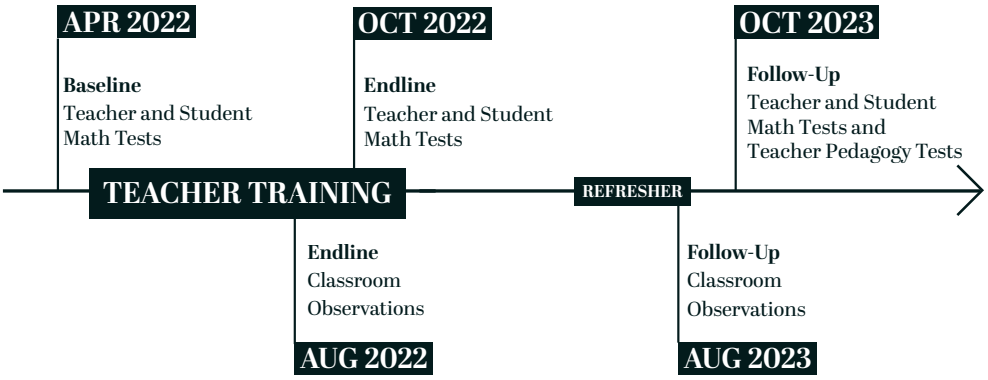


Figure 3: Timeline of the project

tests were conducted at baseline, endline and follow-up, whereas pedagogy tests were only implemented at follow-up (see Figure 3). We record a total of 338 teacher test observations at baseline, 309 at endline, and 297 at follow-up.

All **math assessments** shared a consistent structure and had a duration of 90 minutes. Each test comprised 40 grade two to grade six items[8], 20 of which have an overlap with the student math assessments. The content of the tests was designed to be closely aligned with the composition of the curriculum, meaning that it was composed of $\sim$ 70 percent NSEA (numeric sense and arithmetics), $\sim$ 25 percent geometry and $\sim$ 5 percent DSP (data, statistics and probability) questions. To avoid errors in grading, the assessments were reviewed by two different people and inconsistencies resolved accordingly.

The **pedagogy assessment** was divided into two sections, spanning a total of 60 minutes. The *first, main part of the test* consisted in designing a lesson plan for a fifth-grade math class. Teachers were presented with a fairly detailed description of a class that has a wide age and ability range. Their job was to develop a lesson plan for a 45 minutes math class on the topic of adding and subtracting fractions. The task description included details on students previous knowledge of the topic and general guidelines for the layout of the lesson plan. Instructions advised teachers to dedicate about 50 of the 60 total minutes to this task. The *second, smaller sub-task* involved giving suggestions on how to manage and improve classroom behavior in a hypothet-

[8]Second and third grade items together have the same occurrence weight as the items from each of grades four, five, or six individually. Teacher items that intersect with student tests were picked from the general student item pool (as opposed to the student item pool of the specific wave).

ical setting of a fifth-grade math class, where students are restless and disruptive. The remaining 10 minutes of the test were reserved for this task. To minimize subjectivity in the grading of the tests, they were corrected twice and averaged where the outcome was a numeric, likert-scale value and re-evaluated where the outcome was binary and coders disagreed. We use principal component analysis (PCA) to create a standardized index for pedagogical knowledge based on the evaluation criteria for the pedagogical assessment. The eigenvalue distribution and corresponding scree plot shown in Figure A3 clearly meet the psychometric literature's criteria for unidimensionality. For robustness, we also present a weighted mean score of the evaluation criteria.[9] Lastly, Table A11 also presents the results by grouping evaluation criteria of the pedagogical assessments into several sub-indicators: exam completeness, overall quality, pedagogy score, lesson plan structure, practical planning score (materials, timing, learning objectives) for part 1 of the exam as well as the number of measures and the quality of measures for part 2.

To shed light on potential mechanisms, as one of our secondary outcomes, we conduct **classroom visits**. Classrooms were monitored at endline and follow-up (see Figure 3). We record a total of 309 observations at endline and 231 at follow-up. Observations were completed using the Stallings instrument, which captures classroom snapshots at regular intervals throughout the lesson (Stallings, 1973). The seven snapshot record what the teacher is doing, what materials she or he is using, and how many students are involved in the activity. For students not involved in the teacher's activity, the recording also documents activity, material and number of students engaged. The instrument classifies as low-inference as the snapshots do not rely on the observer's personal interpretation of what is captured. As such, it yields a reliable and objective estimate of the use of the teacher's time and materials. To better link the instrument to our pedagogical interventions, we further incorporate a short set of pedagogical questions targeted at basic concepts of the pedagogical teacher training such as whether there was a lesson outline or summary and whether the teacher reviewed or gave homework.

Lastly, we collect detailed **survey and program implementation data**. To shed light on possible channels of impact, teachers were asked both at endline and follow-up about their perceived impact on students' learning, math lesson preparation,

[9]The average standardized score for part 1 of the assessment was weighed 5/6 and the average standardized score for part 2 was weighed 1/6. Criteria for part 1 were all on a scale from 1 to 3 (No, Somewhat, Yes), whereas part 2 was evaluated using the number of classroom management measures suggested and whether measures were deemed appropriate on a scale from 1 to 4. For part 2, criteria were first standardized, then averaged.

motivation and perceived competence to give a math lesson. If they taught subjects other than just math, they were additionally asked to draw comparisons to math across these aspects. Since surveys were administered immediately after teacher math assessments, the number of observations (338 at baseline, 309 at endline, and 297 at follow-up) align. To track the general intervention implementation and compliance, we keep thorough logs of teachers' attendance of on-site workshops, online meetings and self-study module completion. At endline, we also conducted semi-structured interviews with participating teachers to gather feedback on the training, identify implementation challenges, and collect suggestions for improvement.

2.3.2 Student Data

As the second main outcome of our study, we administer math tests to students at baseline, endline and follow-up. This allows us to evaluate whether potential learning gains in teacher knowledge translate into enhanced student performance. Exams were accompanied by a short survey that asked students for background data, their motivation and their favorite and least favorite subject as well as activity. We record a total of 6,010 student exams and surveys at baseline, 4,950 at endline and 2,441 at follow-up.

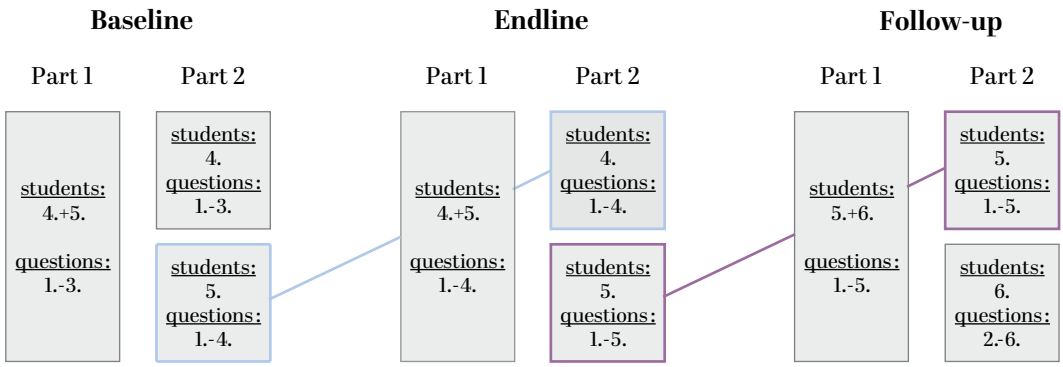


Figure 4: Assessment structure

Notes: Structure of the student assessments for each data collection wave. The numbers refer to the respective grade levels covered by the tests. Blue and purple lines illustrate assessment overlap.

Student exams followed a structure comparable to that of the teacher assessments. Consistent with the teacher assessments, student tests included 40 questions. Due to the lower complexity of the items, they only lasted 60 minutes. For the estimation of the joint IRT model, items of each grade overlapped within each wave as well as

over time. The first part of the test consisted of 20 items solved by all the students regardless of their grade. In a second part, tailored to each grade, students had to solve 20 more questions. As this part was grade-specific, it could be repeated in the subsequent data collection waves for the corresponding grade (see Figure 4 for a graphical representation of the exam structure). Just as with the teacher exams, the student assessments were designed to reflect the Salvadorian math curriculum in terms of topics.[10]

2.4 Item Response Theory: Linking Waves, Students, and Teachers

We leverage the overlap between tests to estimate a joint Item Response Theory model using all teacher and student data from all waves. IRT provides a model-based alternative to percent-correct scoring by explicitly linking observed responses to an underlying latent ability θ. Unlike raw scores, which assume all items carry equal information, IRT accounts for systematic differences in item difficulty, discrimination, and (for multiple-choice items) the probability of guessing. This allows us to place teachers and students on a common scale, harmonize student scores across waves, and ensure comparability over time.

To improve estimation efficiency and model convergence, we employ a mixed IRT specification: we estimate discrimination and difficulty for all items, but allow the guessing parameter to be freely estimated only for multiple-choice questions (3PL; about 20% of the item pool), while fixing it to zero for open-response items (2PL). The correlation between ability estimates from this mixed model and those from alternative specifications like a full 2PL or 3PL model exceeds 99.5%, indicating that results are only minimally affected by this modeling choice.

As detailed in Chapter 2.3.2 and illustrated in Figure 4, part one of the student exam provides the within-wave link, while part two provides the across-wave link. Additionally, half of each teacher's test consists of items also answered by students, allowing us to jointly scale teachers and students.

Our model validation proceeds in several steps. First, we test for differential item functioning (DIF) across grades and between teachers and students. As reported in Appendix A6, very few items exhibit DIF. Second, only three items show discrimina-

[10]The tests covered between three and five grade levels. Since students at baseline were only starting their school year, they were tested on material up to the previous grade level. In general, the distribution of items placed greater weight on higher grade levels closer to students' actual grade.

tion parameters below standard thresholds.[11] Moreover, we check Cronbach's alpha for each wave and for the across-wave link to confirm internal consistency.[12] Third, excluding items with DIF or low discrimination yields IRT scores that correlate at 99.99% with our baseline estimates, further demonstrating the stability of our results. Lastly, we construct a Wright (item–person) map (see Figure A7) that plots item difficulties against the distribution of latent ability. The figure shows that item difficulties are concentrated in the upper two-thirds of the ability range, yielding high information in that region but limited precision for the lowest third of students. Estimates in the lower tail therefore depend more on extrapolation than on direct item-level variation and are thus less tightly identified.

2.5 Balance at Baseline and Attrition

Table 1 confirms that randomization successfully achieved balance across experimental groups. Only one of our baseline variables for teachers and students—teacher baseline scores—shows significant differences between groups. Although teacher baseline performance was used as a stratification variable to improve balance across treatment arms, some within-stratum imbalance remains. This imbalance drives the bivariate orthogonality tests, producing differences between the control group and both T1 ($p = 0.14$) and T2 ($p = 0.08$). When teacher baseline scores are excluded from the balance checks, we no longer come close to rejecting equality across groups. To address this, we include teacher baseline scores as controls in all empirical specifications.

The average teacher is 47 years old with approximately 12 years of teaching experience, though both age and experience vary widely, ranging from 22 to 70 years and from 0 to 45 years of experience. The majority, 62% of the sample, are women and about 90% of the sample teaches other subjects in addition to math. Around 40% of teachers instruct math in both fourth and fifth grade, and the average class size is 16 students. On average, teacher answered about 61% of the baseline math items correctly (for a more detailed analysis of teacher and student baseline performance see Section 3.1).

Student performance at baseline was relatively low, with an average score of 29%. Math is the favorite subject for about 40% of students, while another 40% rank it among their least favorite. The sample is evenly split between male and female

[11] Items with discrimination below 0.34 are widely viewed as weakly informative (Baker, 2017).

[12] A Cronbach's alpha of ≥ 0.8 is commonly interpreted as evidence of good internal consistency (Nunnally, 1994; Bland & Altman, 1997; DeVellis, 2016). Cronbach's alpha indicates high internal consistency within waves ($\alpha = 0.85$–0.93) and acceptable to good reliability for the across-wave linking items ($\alpha = 0.76$–0.80).

students, and the average student is about 11 years old. The typical student lives in a household with five members, including the student. The socioeconomic status is measured using a principal components index constructed from binary indicators for household assets, including electricity, a TV, a computer, a fridge, internet access, a car, a washing machine, and books, which is why its mean is not interpretable. On average, households possess approximately 70% of the listed assets.

Table 1: Balance at baseline

	Control	T1: Math	T2: Pedagogy	T3: Combined	P-value	N
Teacher variables						
Baseline score	0.59	0.61	0.63	0.61	0.04	338
	(0.20)	(0.22)	(0.22)	(0.21)		
Sex	0.64	0.62	0.59	0.64	0.86	338
	(0.48)	(0.49)	(0.50)	(0.48)		
Age	45.60	47.90	46.86	46.14	0.42	336
	(10.36)	(10.82)	(9.63)	(10.12)		
Class variables						
Class size	12.95	14.16	12.78	13.30	0.60	333
	(6.55)	(7.02)	(7.05)	(6.58)		
Student variables						
Baseline score	0.29	0.29	0.28	0.30	0.30	6010
	(0.16)	(0.16)	(0.15)	(0.16)		
Favorite subject math	0.43	0.41	0.42	0.42	0.88	5399
	(0.49)	(0.49)	(0.49)	(0.49)		
Female	0.49	0.50	0.51	0.49	0.44	5684
	(0.50)	(0.50)	(0.50)	(0.50)		
Age	10.62	10.67	10.71	10.68	0.75	5384
	(1.01)	(1.12)	(1.04)	(0.96)		
SES index	0.14	0.02	-0.01	0.05	0.48	5249
	(1.39)	(1.43)	(1.52)	(1.54)		
Household members	5.33	5.33	5.26	5.25	0.76	5705
	(2.14)	(2.22)	(2.17)	(2.15)		
Orthogonality Tests						
Bivariate: Control vs Math					0.137	
Bivariate: Control vs Pedagogy					0.075	
Bivariate: Control vs Combined					0.630	

Standard errors in parentheses clustered at the teacher level. Individual p-values are from F-tests that treatment status does not predict each individual outcome variable, calculated controlling for age, sex, and stratum fixed effects, except for age (does not include age fixed effects) and sex (does not include sex fixed effects). Bivariate orthogonality tests examine whether baseline variables predict treatment assignment in pairwise comparisons between each treatment arm and the control group. Orthogonality tests are computed using randomization inference (1000 replications) which mitigates concerns about over-rejecting when many balance variables are included (see (McKenzie, 2024).) Orthogonality tests include stratum fixed effects. All standard errors are clustered at the school level.

Table 2 reports attrition rates for teacher assessments, classroom observations,

and student assessments across data collection waves. Attrition at endline was low and balanced across treatment arms, amounting to 9 percent for teachers (both assessments and classroom observations) and 18 percent for students. At follow-up, attrition remained modest for teacher assessments (12 percent) but increased substantially for student assessments (59 percent) and classroom observations (31 percent).

A large share of follow-up attrition reflects structural ineligibility rather than survey non-response. In the student sample, students were only re-assessed if they continued to be taught by a sampled teacher. Consequently, teacher reassignments, changes in teaching responsibilities, and promotions to administrative positions generated substantial mechanical attrition among students even when the teachers themselves remained in the study. Between endline and follow-up, 123 teacher clusters ceased to teach the original study cohort, resulting in the loss of approximately 2,700 students or 45% from the follow-up sample. The same institutional features affected classroom observations. Because classroom observations required teachers to be actively teaching mathematics at follow-up, 14% of teachers became structurally ineligible for observation after no longer teaching mathematics or moving into administrative roles. Attrition in classroom observations was nevertheless lower than in the student sample because observations did not require teachers to remain with their baseline class and could be conducted in any mathematics class between grades 3 and 9 when no grade 4 or 5 class was available.[13] Despite these challenges, we find no evidence of differential attrition across treatment arms that would threaten the internal validity of our estimates (Table A1; see also Tables A4 and A5). Moreover, despite substantial structurally induced attrition at follow-up, we find no evidence of selective attrition. Although conventional attritors at both endline and follow-up exhibit somewhat lower baseline achievement and are, on average, older than non-attritors (Tables A2 and A3), structurally ineligible attritors do not differ systematically from non-attritors along observable baseline characteristics in either classroom observations or student assessments (Table A3).

[13]We consider whether the loss of statistical power stemming from attrition at follow-up may constrain inference. We therefore revisit the original power calculations reported in 2.2 and reestimate minimum detectable effects (MDEs), adjusting the number and size of clusters to the realized follow-up sample while holding all other parameters fixed. These revised estimates indicate that, despite substantial attrition, the study remains sufficiently powered to detect effects of 0.153σ–0.162σ with 80 percent probability at the 5 percent significance level.

Table 2: Attrition per data collection type and wave

	Attrition Rates			*Number of Observations*	
	Endline	*Follow-up*		*Endline*	*Follow-up*
	Conventional Attritors	Structurally Ineligible	Conventional Attritors		
Teacher tests	0.09	0.00	0.12	309	297
Classroom observations	0.09	0.14	0.17	309	231
Student tests	0.18	0.45	0.14	4950	2441

Attrition rates for teachers at follow-up differ between classroom observations and teacher assessments because teacher tests were administered to all sample teachers, whereas classroom observations were only feasible if teachers were still teaching mathematics (grades 3–9) at follow-up. Student attrition at follow-up is mechanically high because students assigned to a different mathematics teacher were no longer part of the study sample and therefore not re-tested. Follow-up attrition is therefore split into structurally ineligible attrition and conventional attrition.

3 Results

3.1 Descriptive Results

Panel (a) in Figure 5 presents the results of the baseline math assessments for teachers. The average teacher is able to correctly answer 61% of grade two to six questions. Teachers' highest performance was observed in basic numeric operations and more mechanical tasks like subtraction with two digits and fraction addition, where they achieved around 70%–90% correct responses. Similarly, they performed well in basic geometric tasks like converting units or reading time (∼90%). However, their performance declined considerably in more complex concepts requiring higher-order reasoning. For instance, only 40% correctly calculate a simple percentage, less than a third understands the order of operations (30%), or how to determine the volume of a cube (31%) and only about one out of six can solve a simple inverse proportionality problem (17%). Panel (a) from Figure 6 further illustrates this pattern, showing a sharp decline in correct answers as question difficulty increases beyond grade three. On average, teachers solved fewer than half of the grade five questions correctly (49%) and only about a third of grade six questions (38%). The observed knowledge gaps suggest that the majority of teachers lack a firm command of the primary curriculum, leaving many in the difficult position of having to teach material they have not fully

mastered themselves.

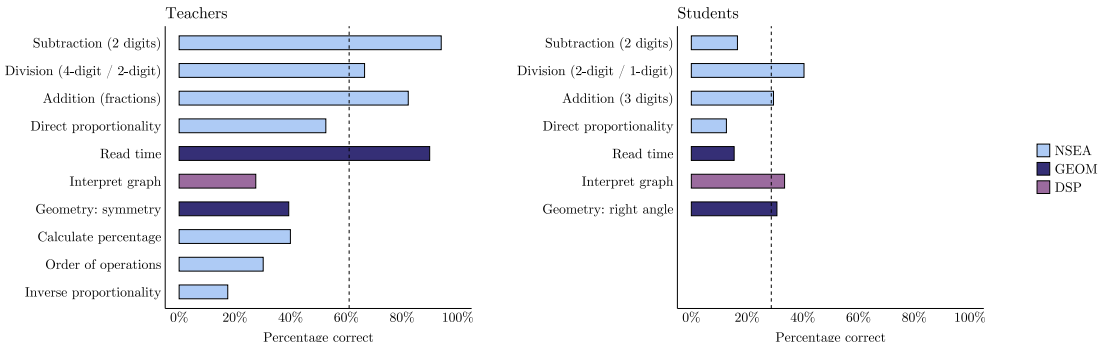


Figure 5: Average percentage per item examples

Notes: The two figures compare mean performance on selected baseline assessment items across all treatment groups. The figures present teacher results on the right and student results on the left. On average, teachers scored 61% correct and students 29%.

Panel (b) in Figure 5 paints a similarly concerning picture for fourth and fifth grade students, with even lower overall performance. On average, students correctly answered only 29% of the math baseline questions, which assessed grade one to three content aligned with the curriculum. As with teachers, students performed better on the simplest tasks—such as counting from 1 to 10 or solving basic one-digit arithmetic—where accuracy ranged between 50% and 90%. However, performance dropped sharply for slightly more advanced problems. For example, only one out of four students knows how to solve an addition problem involving a three-digit number, one in five correctly divides a two digit number by a one digit number, and just one out of six students knows how to subtract two digit numbers from each other. Fifth graders slightly outperform fourth graders (see Panel (a) in Figure 6), suggesting that an additional year of schooling helped reinforce at least some foundational skills. However, the decline in performance as item difficulty increases is even steeper for students than for teachers, pointing to a clear deficiency in fundamental math competencies that are critical for progression in higher grades.

By concurrently calibrating the IRT model with both teacher and student responses, we are able to directly compare students and teachers latent traits at baseline. Panel (b) from Figure 6 illustrates the distribution of these traits across groups. While teachers, as expected, generally outperform students, the lower tail of the teacher distribution nevertheless overlaps with the student distribution. Approximately 57 percent of students outperform the lowest-performing teacher, while approximately 6 percent outperform teachers in the bottom decile of the teacher distribution. The

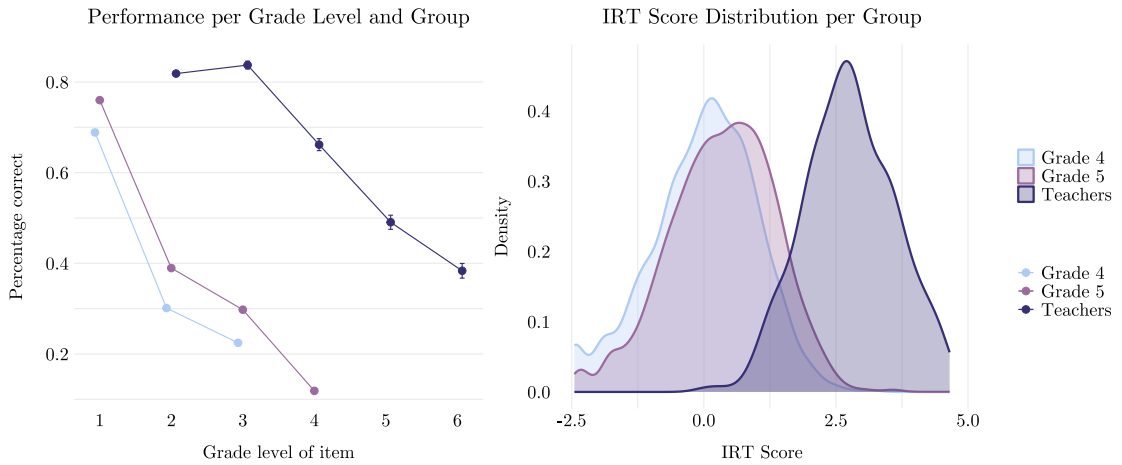


Figure 6: Average baseline performance per group

Notes: The left figure shows percent correct per grade level of the items for teachers and students. Since the baseline assessment took place at the beginning of the school year, Grade 3 items correspond to grade-level content for fourth graders, while Grade 4 items correspond to grade-level content for fifth graders. The right figure illustrates the distribution of baseline irt scores for each teachers and students. IRT scores are standardized to the fourth grade baseline distribution.

overlap, however, is primarily driven by variation across clusters, as only 72 out of roughly 6,000 students outperform their own teacher. This descriptive comparison complements the broader literature linking teacher knowledge to student achievement (e.g., Metzler and Woessmann, 2012; Bold et al., 2019; Bau and Das, 2020) and suggests that student ability may be largely constrained by teacher ability—a student can only achieve as much as their teacher enables them to.

To place Salvadorian students' and teachers' abilities in an international context, we benchmark their performance to the TIMSS scale, building on the IRT-based linking framework proposed by Patel and Sandefur (2020) (see Appendix A5.2). Figure 7 illustrates this international comparison, reinforcing our earlier descriptive findings. The TIMSS test in 2015 was conducted with fourth graders from 49 countries, most, but not all, of which are high-income countries. Since TIMSS is administered later in the academic year than our assessments, the true correspondence between our fourth- and fifth-grade scores and the TIMSS benchmark plausibly lies between the two grades. El Salvador's fourth and fifth graders rank near the bottom of the international distribution, positioned just above Kuwait, the lowest-performing country in the sample. Among the nine middle-income countries that took part in the TIMSS assessment, El Salvador scored lowest. Even more strikingly, El Salvador's teachers only very narrowly outperform Singapore's fourth graders—the highest-scoring students in

the assessment. That teachers struggle to surpass 10-year-olds from top-performing systems is yet again a staggering reflection of the profound knowledge gaps within El Salvador's education system.

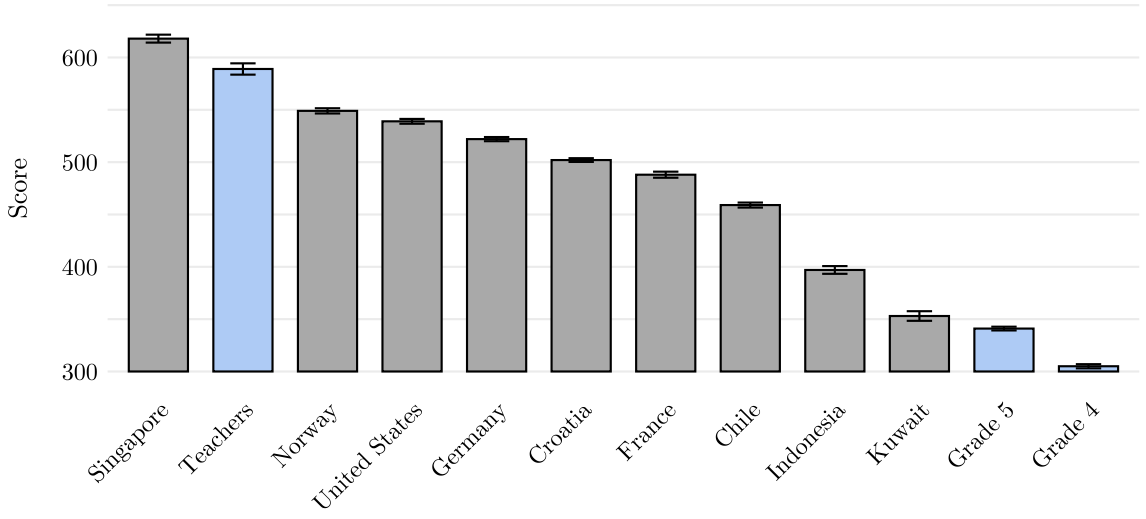


Figure 7: El Salvador in international comparison: TIMSS

Notes: The blue bars represent teachers' and students' test scores mapping from El Salvador using our own data. The TIMSS test in 2015 was conducted with fourth graders from 49 countries, most of which are high-income countries. Among these countries, Singapore achieved the highest scores, while Kuwait scored the lowest. The figure highlights additional countries that provide meaningful points of comparison.

3.2 Experimental Results

To assess the causal effect of the three treatments on teacher and student outcomes for each post-treatment $wave \in \{\text{endline}, \text{follow-up}\}$, we estimate

$$Y_{ic}^{wave} = \sum_{k=1}^{3} \beta_k T_k + \delta \tilde{Y}_{ic}^{baseline} + \sum_{k=1}^{3} \gamma_k \left(T_k \times \tilde{Y}_{ic}^{baseline} \right) + \mu_s + \epsilon_{ic}, \tag{1}$$

where Y_{ic}^{wave} denotes the endline or follow-up outcome for student i taught by teacher c; T_1, T_2, and T_3 are indicators for the math, pedagogy, and combined treatment arms, respectively; $\tilde{Y}_{ic}^{baseline}$ is the demeaned baseline outcome; and μ_s are strata fixed effects. Following Lin (2013), baseline outcomes are demeaned and interacted with each treatment indicator in all specifications. Standard errors are clustered at the teacher level. For teacher outcomes, we additionally control for sex and age. For student outcomes, we additionally control for teachers' baseline scores to account for

baseline differences across treatment groups (see Section 2.5). As an alternative specification, we also control for student sex and age, imputing missing values with the control-group mean and including missing-value indicators.

Table 3 presents the estimated impacts on student learning outcomes immediately after the program and one year later. None of the three treatment groups exhibit statistically significant differences in math scores relative to the control group at either point in time. The small point estimates and standard errors suggest that the true effects are close to zero. Even the largest positive estimate across treatments, waves, and outcome specifications (0.029σ, corresponding to the endline specification with controls using percent-correct math scores) allows us to rule out effects above approximately 0.16σ at the 95% confidence level.[14]

When examined separately by grade, the immediate post-intervention effects show slightly positive point estimates for fourth-grade students and slightly negative ones for fifth-grade students (Table A10). We find no corresponding heterogeneity in teacher knowledge, classroom implementation, or teacher beliefs and attitudes, and are unable to identify a plausible mechanism through which all three interventions would reduce learning specifically among fifth-grade students. Furthermore, the pattern does not persist at the one-year follow-up. Taken together, we interpret these grade-specific estimates with caution.

4 Discussion

Our theory of change posits four critical intermediary steps necessary for improving student test scores. *In a very first step*, teachers must attend the training and the program must be implemented with fidelity. *Second*, teachers must achieve meaningful learning gains to have the requisite knowledge and skills to transmit to their students. This also implies that improvements in teacher knowledge, be it in content or pedagogical knowledge, must not only be detectable but also substantial enough to influence student learning. *Third*, particularly in the case of pedagogical competencies, but also for content knowledge through explanatory quality, feedback and pacing, these newly acquired skills must be successfully integrated into classroom instruction. *Fourth*, even if teachers successfully enhance knowledge and their practice, student learning outcomes will only improve if students can effectively absorb, pro-

[14] As a robustness check against the differential follow-up attrition discussed in Section 2.5, Appendix Table A13 re-estimates the endline effects using only students who were subsequently observed at follow-up. The estimated treatment effects remain similar, suggesting that differential follow-up attrition is unlikely to affect our conclusions.

Table 3: ITT estimates: Student math scores

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Without Controls								
Math: Share correct responses (%)	0.028	0.001	-0.097	4950	0.005	-0.007	-0.033	2441
	(0.065)	(0.076)	(0.062)		(0.079)	(0.083)	(0.095)	
Math: IRT score (sd)	0.009	-0.043	-0.094	4950	0.017	-0.038	-0.024	2441
	(0.062)	(0.072)	(0.061)		(0.077)	(0.085)	(0.093)	
Math: TIMSS score (points)	0.910	-6.208	-7.960	4950	2.170	-4.998	-1.959	2441
	(5.535)	(6.392)	(5.576)		(7.060)	(7.905)	(8.506)	
With Controls								
Math: Share correct responses (%)	0.029	-0.002	-0.098	4950	0.011	0.001	-0.024	2441
	(0.065)	(0.076)	(0.062)		(0.077)	(0.082)	(0.094)	
Math: IRT score (sd)	0.010	-0.046	-0.095	4950	0.023	-0.030	-0.015	2441
	(0.061)	(0.072)	(0.061)		(0.075)	(0.083)	(0.092)	
Math: TIMSS score (points)	0.974	-6.542	-7.990	4950	2.686	-4.214	-1.046	2441
	(5.519)	(6.381)	(5.600)		(6.947)	(7.790)	(8.425)	

Controls include strata fixed effects, the baseline value of the outcome, student sex, age, and teacher baseline score. Missing values for age and sex are replaced with the control group mean, with missing-value indicators included in the regressions. Standard errors are clustered at the teacher level. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

cess, and retain the newly introduced content. The extent to which these conditions are met determines whether the intervention leads to measurable gains in student achievement.

4.1 Compliance and Implementation

We first show that our intervention was implemented with high fidelity and avoided the early compliance and delivery failures that derailed comparable government programs and blocked the first and necessary step of improving teacher knowledge and classroom practice (Loyalka et al., 2019; Schaffner et al., 2025).

Table 4 summarizes program participation. Over all treatment groups, 81% of teachers attended at least one of the seven in-person workshops with the average teacher completing 4.75 out of the seven workshops (including non-participants). Participation rates for online self-study modules and required homework submissions were similarly high (see Table 4). Consistent with this pattern, teachers in every treatment arm reported significantly greater perceived workload, rising by roughly 0.5σ relative to the control group ($p < 0.01$; Table A6). Participation is slightly lower in the combined and pedagogy treatments relative to the math group, as reflected in the number

of completed workshops ($p < 0.01$ and $p = 0.11$). This aligns with field reports suggesting that teachers appeared less familiar with the concept of pedagogy and were primarily motivated to participate by the availability of math training. Participation remained stable over time, with no clear patterns or variations across all treatments (see Figure A4).

Table 4: Compliance by treatment

Treatment	Attended one or more (%)	Completed workshops	Completed online activities	Completed homework sets
T1: Math	0.85	5.36	5.06	5.59
T2: Pedagogy	0.82	4.69	4.46	5.08
T3: Combined	0.77	4.18	4.15	4.35

The total number of activities for each block is 7.

High participation rates imply that non-compliance is unlikely to explain the absence of student learning gains. To assess this directly, we estimate complier average causal effects using a 2SLS specification that instruments attendance at a majority of workshops (four of seven) with treatment assignment. As shown in Table A12, the resulting estimates remain small and statistically insignificant across treatment arms, outcome measures, and survey rounds, suggesting that incomplete take-up is not the primary explanation for the null results.

Loyalka et al. (2019), the first of two identified studies that assess classical government training programs in primary schools but find null effects for students and teachers, attribute the failure of their program to content that was overly theoretical and delivered in a rote, non-interactive manner. In designing our intervention, we explicitly sought to avoid such weaknesses: as described in Section 2.1, the curriculum adhered to rigorous pedagogical standards, was tailored to teachers' knowledge levels, and emphasized interactive learning. A post-training survey of treated teachers suggests that these design features were well received: the median teacher rated the overall quality of the training as 5 ("very good") on a five-point Likert scale. Twelve semi-structured interviews with a randomly selected subset of teachers across treatment arms provide further insight into these perceptions. Teachers consistently valued the training, highlighted the quality of instruction, and conveyed a clear sense that the sessions differed positively from traditional government trainings (see Figure A10 for exemplary quotes from the interviews). Many also provided concrete examples of how the interactive methods altered their classroom practice. When asked how the training could be improved, teachers generally offered no suggestions and replied that

"everything was great".[15] Instead, nine of the twelve interviewed teachers expressed a desire for the program to continue, either by extending it over time, expanding it to additional teachers, or applying the same training model to other subjects.

Schaffner et al. (2025), the second study on classical government training, identify low instructor motivation, inadequate preparation and support for instructors, and weak subject knowledge of instructors as key barriers to generating meaningful teacher and student impacts. To address these concerns, our intervention invested heavily in instructor selection and support. Facilitators were young teachers hired by the implementing NGO, each with at least a bachelor's degree in mathematics or educational science (3-6 years oof university instruction) and relevant classroom experience. Of seventeen pre-screened candidates, twelve were ultimately hired following subject-matter testing and 1.5 weeks of initial training. Upon hiring, they received two additional weeks of general preparation as well as block-specific training delivered by the respective Salvadorian expert ahead of each instructional block. Their schedule also allowed roughly two weeks of preparation time per block. During the intervention, facilitators received weekly in-person and online coaching from Salvadorian and Swiss experts and two monitoring visits to ensure quality. According to NGO reports, all twelve facilitators demonstrated strong motivation and performance throughout.

4.2 Teachers' Learning Gains

Unlike Schaffner et al. (2025), who also posit that, aside from instructor-related issues, their training did not target teachers' actual knowledge levels, we find that our intervention does so and leads to significant gains in teacher knowledge. Table 5 shows effect sizes for teachers' content and pedagogical knowledge. Immediately after the intervention, teachers in the math treatment group improved their scores by 0.14σ ($p < 0.05$). These effects increased to 0.29σ ($p < 0.01$) one year later, suggesting that teachers continued consolidating their knowledge over time. This effect is sizeable, corresponding to a 23-point shift on the TIMSS scale. It is comparable to the gap between Salvadorian fourth and fifth graders and to the difference between the fourth-grade country averages of Chile and France. As expected, there were no significant impacts on math scores for teachers who received only the pedagogy training.[16]

[15] A word cloud of responses to the improvement question shows that "everything" was the most frequently mentioned term, and it appeared only alongside positive descriptors such as "great" and "amazing" (Figure A13).

[16] Point estimates for the pedagogy treatment are negative at endline but imprecisely estimated. We see no credible mechanism through which pedagogy training would reduce teachers' mathematical knowledge, and the corresponding follow-up estimates are positive. Teacher survey responses also

For teachers in the combined training group, we observe an initial improvement of 0.11σ relative to the control group, although the effect falls just short of statistical significance ($p = 0.13$). Similar to the math-only group, the effect increases to 0.15σ in the follow-up ($p < 0.1$), indicating that these teachers also strengthened their math skills over time.

Table 5: ITT estimates: Teacher scores

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Teachers: Math								
Math: Share correct responses (%)	0.144**	-0.107	0.109	309	0.293***	0.041	0.146*	297
	(0.071)	(0.069)	(0.071)		(0.077)	(0.085)	(0.078)	
Math: IRT score (sd)	0.125*	-0.091	0.076	309	0.284***	0.110	0.124	297
	(0.075)	(0.072)	(0.075)		(0.079)	(0.088)	(0.081)	
Math: TIMSS score (points)	11.157*	-6.084	7.970	309	22.942***	10.335	11.630*	297
	(6.525)	(6.259)	(6.859)		(6.619)	(7.122)	(6.760)	
Teachers: Pedagogy								
Pedagogy Score (PCA)	-	-	-	-	0.493***	0.564***	0.469***	295
					(0.168)	(0.182)	(0.179)	
Pedagogy Score (Weighted Mean)	-	-	-	-	0.456***	0.543***	0.433**	297
					(0.165)	(0.177)	(0.177)	

Controls include strata fixed effects, the outcome variable at baseline and controls for teacher age and sex. For pedagogical knowledge, baseline standardized teacher math test scores (percent correct) are included as a proxy since pedagogical knowledge was not measured at baseline. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

We find substantial and statistically significant long-term effects on pedagogical knowledge, as measured by the PCA index, across all three treatment groups. The largest effect, 0.57σ ($p < 0.01$), is observed among teachers who received the pedagogical training. The effects for the math and combined treatments are only slightly smaller, at 0.49σ ($p < 0.01$) and 0.47σ ($p < 0.05$), respectively. This indicates that all three interventions led to lasting improvements in pedagogical knowledge even one year after the program ended. The differences in effect sizes between the pedagogical-training group and the other two treatment groups are not statistically significant ($p = 0.83$). The effects in the mathematics-only treatment likely reflects the training's interactive and pedagogically structured delivery. By experiencing effective instructional practices while learning mathematics, teachers appear to have

provide no evidence of greater self-reported difficulty teaching or preparing mathematics lessons (Table A6 and A7). We therefore interpret the negative endline estimates as likely reflecting sampling variation rather than a meaningful decline in teacher knowledge.

acquired pedagogical knowledge alongside mathematical content, consistent with evidence from our semi-structured interviews. These findings remain robust when using a weighted mean index instead of the PCA-based measure.

For the mathematics treatment, an important remaining question is whether the observed gains in teachers' content knowledge were large enough to translate into measurable student learning. While effects for teachers in the math training reach 0.27σ for specific math domains (see Table A14) and 0.29σ at follow-up, the implied improvement in student learning may nevertheless be modest. Quasi-experimental evidence from Peru (Metzler and Woessmann, 2012), several African countries (Bietenbeck et al., 2018; Bold et al., 2019), Pakistan (Bau and Das, 2020), and El Salvador (Brunetti et al., 2024) suggests that a 1σ increase in teacher content knowledge is associated with only about a 0.1σ increase in student learning outcomes. Our own data corroborate this relationship, suggesting that a 1σ increase in teacher content knowledge is associated with a 0.08σ increase in student performance (see Table A16). Under this benchmark, the improvements in teacher knowledge observed in our study would translate into expected gains of roughly 0.015-0.03σ, which our study is unlikely to be powered to detect. However, this benchmark is likely conservative in our setting because the mathematics training also generated substantial improvements in teachers' pedagogical knowledge. As a result, the intervention's impact on student learning may not be fully captured by teacher–student knowledge transmission patterns documented in the existing literature.

4.3 Classroom Implementation

Given these gains in teachers' pedagogical content knowledge, we next examine whether they translate into changes in classroom instruction. We find that teachers integrate newly acquired pedagogical skills into classroom instruction in the short run. Specifically, we observe a shift in classroom practices at endline for the treatment groups that received a pedagogical component in their training (see Table 6[17]). Teachers in these treatment groups are 4–5 percentage points more likely to use interactive materials, such as shared resources, learning aids, and information and communication technology, than teachers in the control group ($p < 0.05$). This corresponds to a 110–130% increase relative to the control mean. At the same time, their reliance on conventional materials, such as notebooks, textbooks, and blackboards, declines by 8 percentage points ($p < 0.05$).

[17]For the extended table see Table A9.

Teachers in both pedagogical treatment groups also perform significantly better in our constructed pedagogical practices index, scoring approximately 0.32σ higher than their peers in the control group ($p < 0.1$). This index measures instructional quality through six key dimensions, including lesson structuring, assignment and review of homework, lesson summarization, teacher mobility around the classroom, and use of examples. Additionally, teachers in these groups are 18.3 and 12.5 percentage points more likely ($p < 0.01$ and $p < 0.05$, respectively) to implement interactive activities, representing a 200% and 140% increase relative to the control group mean. Improved instructional practices are also associated with enhanced student engagement, as evidenced by a 7.7 percentage point reduction ($p < 0.1$) in classroom distractions for the pedagogy treatment group, amounting to a 20% improvement over the control group mean, while the corresponding estimate for the combined treatment is smaller (4.9 percentage points) and imprecisely estimated.

Table 6: Classroom observations: Teachers

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	Control mean	T1: Math	T2: Ped	T3: Both	Control mean
Teacher use of time								
Active (%)	-0.011	0.002	0.015	0.452	-0.014	-0.059	0.020	0.528
	(0.036)	(0.036)	(0.036)		(0.038)	(0.039)	(0.039)	
Passive (%)	0.006	0.031	-0.023	0.360	0.026	0.048	0.016	0.207
	(0.035)	(0.036)	(0.036)		(0.032)	(0.033)	(0.033)	
Management (%)	0.011	-0.020	0.014	0.132	0.036	0.047	-0.001	0.143
	(0.025)	(0.025)	(0.025)		(0.028)	(0.029)	(0.029)	
Material use								
Interactive Material (%)	0.039	0.053**	0.052**	0.038	0.030	0.028	0.032	0.043
	(0.024)	(0.024)	(0.024)		(0.025)	(0.026)	(0.025)	
Traditional Material (%)	-0.056	-0.077**	-0.088**	0.748	-0.057	-0.052	-0.025	0.653
	(0.039)	(0.039)	(0.039)		(0.040)	(0.041)	(0.041)	
Pedagogy								
Pedagogical score (sd)	0.105	0.320*	0.300*	0.000	-0.143	0.127	-0.062	0.000
	(0.168)	(0.170)	(0.171)		(0.156)	(0.161)	(0.160)	
Activity variation (sd)	0.178	-0.024	0.061	0.000	0.177	-0.056	0.014	-0.000
	(0.142)	(0.143)	(0.144)		(0.159)	(0.163)	(0.163)	
Conducts interactive activity	0.007	0.182***	0.118**	0.092	0.096*	0.145***	0.054	0.018
	(0.058)	(0.059)	(0.059)		(0.050)	(0.051)	(0.051)	
One or more students distracted (%)	-0.008	-0.077*	-0.045	0.447	0.008	-0.055	-0.044	0.707
	(0.046)	(0.047)	(0.047)		(0.046)	(0.047)	(0.047)	

Sample sizes are n = 309 for the estimation of endline effects and n = 230 for follow-up effects. All the dependent variables are either standardized (sd), binary or indicated as a percentage of the lesson time (%). Interactive material constitutes shared material, learning aids or ICT. Traditional material consitutes textbook, notebook and blackboard. The pedagocical score is a standardized score following Anderson (2008) of the following variables: Does lesson outline, does homework review, uses examples, moves in classroom, gives homework, does lesson summary. Controls include strata fixed effects and demographic teacher variables.

However, these gains prove to be short-lived. By the one-year follow-up, the improvements in classroom practice have largely dissipated, suggesting that the ini-

tial changes were difficult to sustain once the intervention had ended. One plausible explanation is that maintaining new instructional practices requires continued reinforcement beyond the initial training period. This interpretation is consistent with evidence from our semi-structured interviews, in which teachers repeatedly emphasized the value of the ongoing pedagogical accompaniment they received during the program and identified it as a distinguishing feature relative to other teacher training programs (see Figure A10).

The institutional context may further help explain why such reinforcement is valuable. In many developing countries, teachers work within highly standardized instructional systems and adhere closely to prescribed curricula (Pritchett and Beatty, 2015). This may be particularly relevant in El Salvador, where the government introduced the standardized mathematics textbook ESMATE in 2018. ESMATE provides highly structured lesson plans aligned with the national curriculum and has been shown to improve student learning (Maruyama and Kurosaki, 2024). At the same time, its standardized structure leaves comparatively little room for teachers to deviate from established instructional routines.

Our teacher survey illustrates the central role of ESMATE in classroom instruction. As shown in Figure 8, 90% of teachers report always or very often using the ESMATE lesson plans to prepare mathematics lessons, and virtually all teachers rely on the textbook as their primary instructional guide. Importantly, the intervention did not reduce teachers' reliance on ESMATE, with no evidence of differential usage across treatment groups (Table A7).

Taken together, we interpret these findings as presumptive evidence that teachers revert to their established routines once external reinforcement is removed. This suggests that pedagogical training might be more successful if it is paired with long-term support such as pedagogical coaching that extends beyond training to consolidate initial training gains and prevent regression to pre-intervention teaching practices. By the same logic, training multiple teachers within a school, rather than only one, could create a more supportive environment for implementing new practices across classrooms.

4.4 Student Deficit

As discussed in Sections 4.2 and 4.3, teacher effect sizes and classroom implementation can pose challenges to the successful delivery of teacher training interventions. However, even where these factors are not the primary constraints on student learning, as

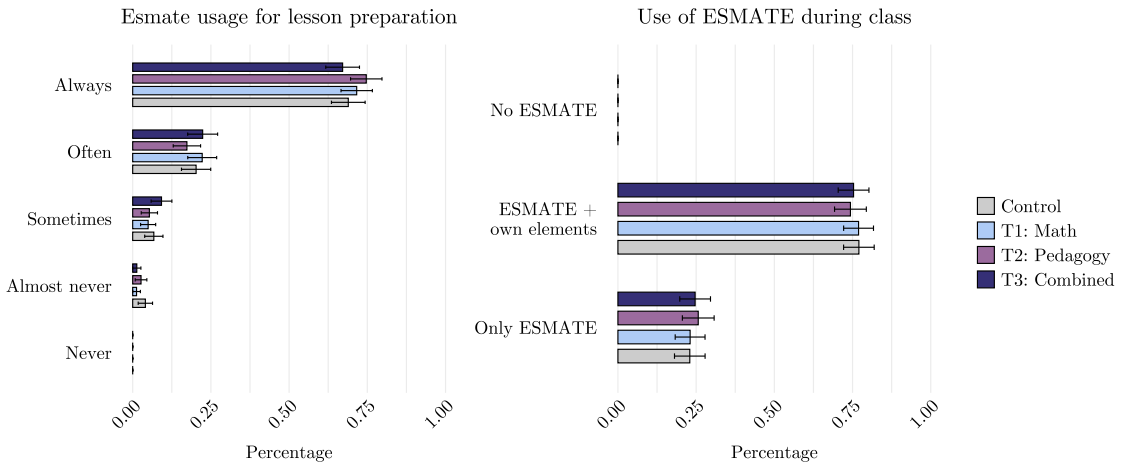


Figure 8: Use of ESMATE during class

Notes: The left figure illustrates how frequently teachers use ESMATE for lesson preparation, while the right figure shows the extent of ESMATE usage during classroom instruction. Both measures are based on self-reported data provided by teachers during the endline data collection. There is no differential use of ESMATE across treatment groups (p-value of a joint F-Test that each treatment dummy coefficient is equal to 0 = 0.98).

suggested by our endline results, students may still face significant obstacles in learning, understanding and retaining the knowledge passed on to them. This can happen if there is a misalignment between the curriculum and their actual competencies.

Section 3.1 illustrates this point. Many students perform well below grade-level expectations: the average fourth grader answers 22 percent of grade-level questions correctly, and the average fifth grader answers 12 percent. Consistent with these patterns, IRT scores place fourth graders at roughly a first-grade level and fifth graders between first- and second-grade levels.

At the same time, evidence suggests that learning trajectories in primary education in developing contexts often follow a concave curve (Patel and Sandefur, 2020; Banerjee et al., 2016). As children progress through grades, their learning gains diminish each additional year, as foundational skills acquired in the early years remain underdeveloped. Once students fall too far behind, they may derive little to no benefit from grade-level instruction (see Muralidharan et al., 2019). Using scores on the TIMSS scale, which are independent of test difficulty, Figure 9 confirms this pattern in our data: average value added per year of schooling declines as cohorts advance through grades within our study period. Students in grade 4 exhibit significantly larger learning gains over the school year than those in grade 5. For cohort 2, learning gains appear to stagnate in grade 6, indicating that the marginal value added of this

grade may be minimal.

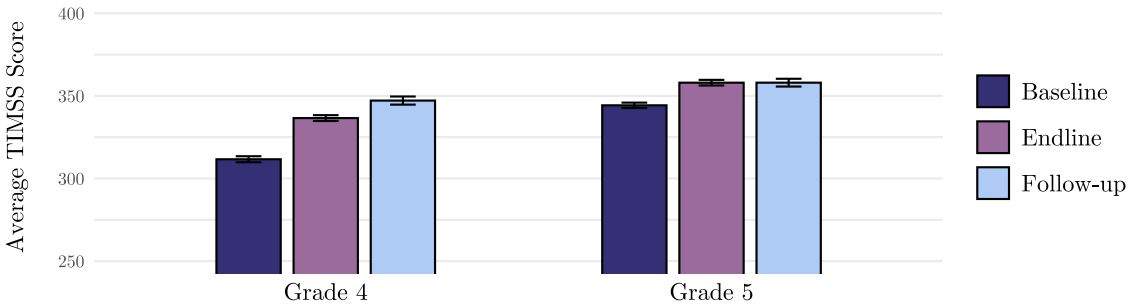


Figure 9: Student learning per grade

Notes: The first cohort corresponds to students who were in grade 4 at baseline and endline and grade 5 at follow-up. Similarly, the second cohort represents students who were in grade 5 at baseline and endline and grade 6 at follow-up. Scores are shown on the estimated TIMSS scale.

Taken together, these patterns suggest that many students may be too far behind the curriculum to benefit from improved grade-level instruction. Evidence from our endline supports this hypothesis. Figure 10 reports treatment effects by students' baseline grade-level deficit, distinguishing between students who were three or more grades behind, two grades behind, and one or fewer grades behind. Grade-level proficiency is defined using the median IRT item difficulty for each grade: students whose estimated ability parameter (θ) exceeds this threshold are classified as having mastered the grade, implying a probability greater than 50% of correctly answering the median-difficulty item. To improve precision, we pool students who were one grade behind with those who were at or above grade level, as only 3% of students met the latter criterion at baseline. Treatment effects are concentrated entirely among these students: those who entered within one grade of proficiency experienced gains of approximately 0.3 standard deviations in both the math and pedagogy arms, whereas students who were two or more grades behind exhibited no meaningful improvements. As a robustness check, Figure A9 reaches the same conclusion using standardized baseline percent-correct scores as the heterogeneity measure in a third-order polynomial specification.

These findings suggest that, in settings characterized by large learning gaps, teacher training programs focused on improving grade-level instruction may face inherent limitations. By contrast, early interventions that strengthen foundational skills may be critical for sustaining learning progress over time and preventing the plateauing of learning trajectories in later grades. This aligns with prior studies, which have

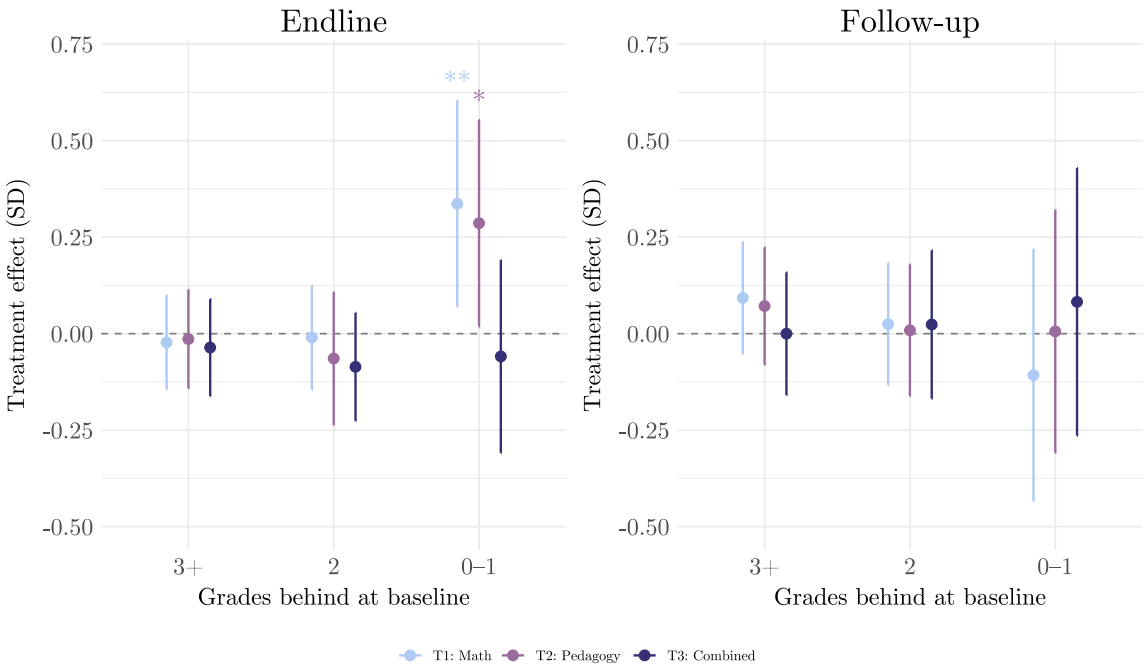


Figure 10: Endline effect heterogeneity over baseline student performance

Notes: Students are grouped by their baseline distance from grade-level proficiency (3+ grades behind, 2 grades behind, and 0–1 grades behind). Grade-level proficiency is defined using the median IRT item difficulty for each grade. The final category pools students who were one grade behind with those at or above grade level. Points denote treatment effect estimates; vertical bars indicate 90% confidence intervals.

predominantly identified positive effects of teacher training in early primary grades, particularly in early reading programs or foundational skill interventions (Kerwin and Thornton, 2021; Piper et al., 2018).

5 Conclusion

Student learning levels remain critically low in many developing countries—a challenge in which teacher quality emerges as a pivotal factor. Using a large-scale randomized controlled trial in El Salvador, we evaluate three intensive teacher training programs targeting subject knowledge, pedagogical skills, or both. The programs succeed where conventional in-service and government-led trainings have fallen short, producing sizable improvements in teacher knowledge and short-term classroom implementation (Loyalka et al., 2019; Schaffner et al., 2025). Yet these improvements do not lead to measurable gains in student learning.

Our results identify two constraints on program effectiveness, even under successful

implementation. First, improvements in classroom practice prove difficult to sustain once the initial training ends. In a context where classroom instruction is guided by highly structured lesson plans, teachers largely reverted to established instructional routines within a year.

Second, effective teaching is limited by a substantial misalignment between the primary curriculum and students' underlying skill levels. At baseline, fourth- and fifth-grade students correctly answer just over 20 percent and 10 percent, respectively, of items at their expected grade level, indicating large cumulative learning gaps. These deficits are also evident in international comparisons, where Salvadorian students rank near the bottom of the TIMSS distribution. In contexts where students lag far below grade-level expectations, even well-trained teachers may struggle to generate sustained improvements if they are expected to teach at their grade level.

These findings have several implications for policy. Sustaining initial gains and preventing reversion to prior teaching practices may require pairing pedagogical training with longer-term support, such as ongoing instructional coaching. This interpretation is consistent with evidence showing that structured pedagogy interventions rank among the most effective education interventions tested (see e.g., Angrist et al., 2025). Relatedly, training only one teacher per school may not be sufficient to instigate widespread change; training multiple teachers within the same school could foster peer interaction and facilitate the diffusion of new teaching practices across classrooms. Finally, and perhaps most importantly, it is crucial to focus teacher training or other educational interventions on mitigating early learning deficits. This is when learning gaps typically first emerge and if left unaddressed, can persistently affect a child's educational trajectory as well as the effectiveness of teacher training programs.

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A Appendix

A1 Literature Review

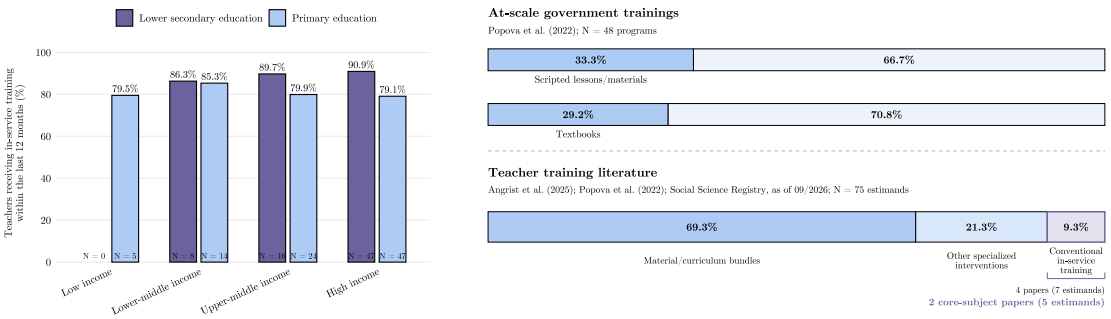


Figure A1: Teacher training in practice and in the literature.

The left panel reports the median share of teachers receiving in-service training in the previous 12 months, by World Bank income group and education level, using the latest available country observation in UNESCO Institute for Statistics SDG indicator 4.c.7; N denotes the number of contributing countries. The right panel summarizes evidence from (Popova et al., 2022; Angrist et al., 2025) and the Social Science Registry (09/2026). The literature sample is restricted to studies since 2000 with at least 30 clusters in primary, secondary, or mixed settings, excluding special education; treatment arms are counted as separate estimands. Interventions were classified by AI based on the pdf and using the prompt reported in the Appendix XX. "Other specialized interventions" include technology or coaching-centered approaches, targeted topics (e.g., HIV, financial literacy, entrepreneurship), and targeted pedagogical approaches (e.g., learning-to-learn, student-centered instruction).

A2 Intervention Design

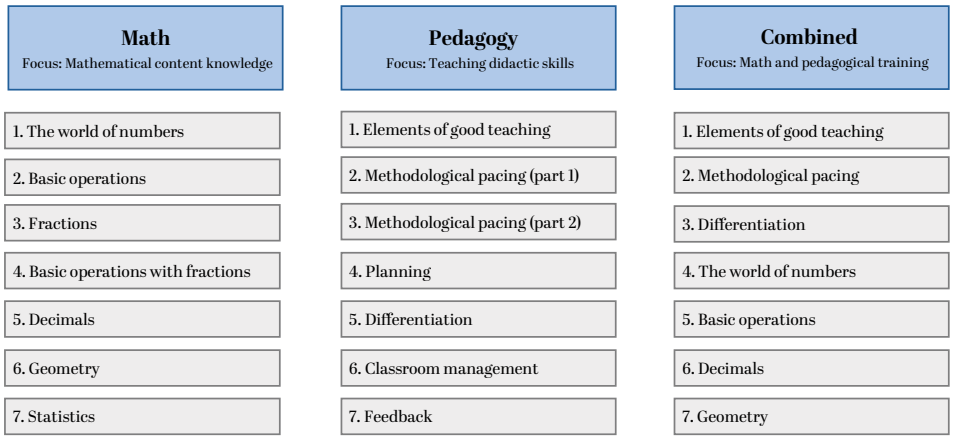


Figure A2: Intervention training block topics

Each training block corresponded to 3 weeks, including one in-person workshop, self-study modules and personalized coaching. See figure 2 for an additional reference.

A3 Data Processing

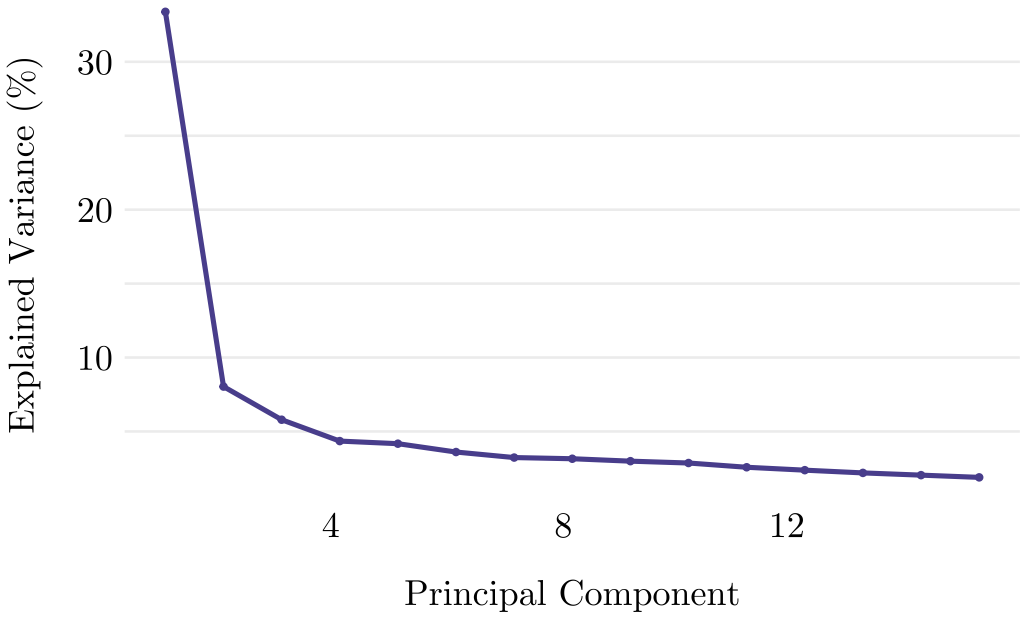


Figure A3: Scree plot

The Scree plot shows the explained variance for each principal component in a factor analysis of all pedagogy evaluation criteria used for the pedagogical teacher assessment at follow-up.

A4 Compliance and Attrition

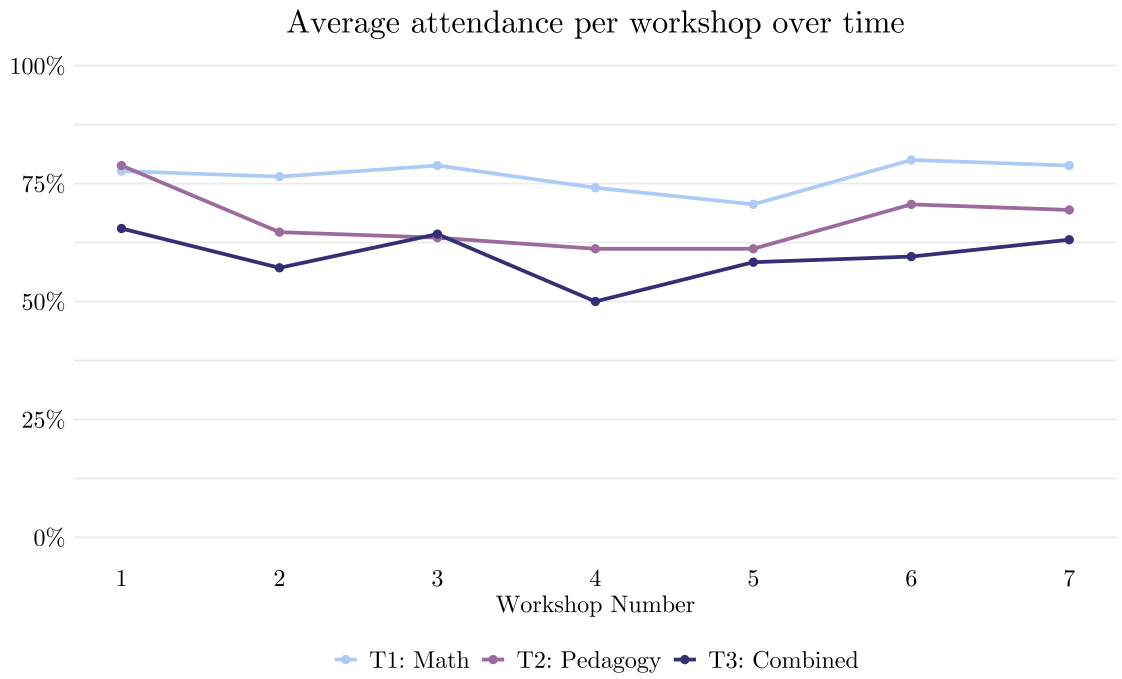


Figure A4: Compliance over time

Table A1: Attrition rates per experimental groups

	Control	T1: Math	T2: Pedagogy	T3: Combined	P-value	N
Endline						
Teacher tests	0.12	0.05	0.11	0.07	0.32	338
Classroom observations	0.10	0.06	0.07	0.12	0.51	338
Student tests	0.16	0.17	0.19	0.19	0.45	6011
Follow-up: Structurally Ineligible						
Teacher tests	-	-	-	-	-	-
Classroom observations	0.17	0.11	0.13	0.18	0.51	338
Student tests	0.45	0.43	0.39	0.54	0.36	6011
Follow-up: Conventional Attrition						
Teacher tests	0.10	0.09	0.14	0.15	0.50	338
Classroom observations	0.17	0.19	0.18	0.15	0.95	338
Student tests	0.13	0.13	0.15	0.15	0.89	6011

The penultimate column indicates the p-value of a joint F-Test that each treatment dummy coefficient is equal to 0. Balance is calculated using strata fixed effects. Standard errors are clustered at the teacher level. Follow-up attrition is split into structurally ineligible attrition and conventional attrition. Structurally ineligible attritors are individuals who could not be re-assessed because they no longer met the study eligibility criteria (e.g. teachers no longer teaching mathematics or students assigned to a different mathematics teacher). Conventional attritors are all other individuals who did not attend the assessment. No structurally ineligible attritors exist at endline or for teacher tests.

Table A2: Attritors vs non-attritors: Endline

Variable	Non-Attritors	Attritors	Difference	P-value	N
Teacher tests					
Baseline score	0.61	0.56	-0.05	0.165	338
Age	46.85	44.18	-2.67	0.130	336
Sex	0.62	0.69	0.07	0.694	338
Classroom observations					
Baseline score	0.61	0.59	-0.02	0.864	338
Age	47.02	42.45	-4.57	0.032	336
Sex	0.62	0.66	0.03	0.942	338
Student tests					
Baseline score	0.29	0.25	-0.04	0.000	6010
Age	10.61	10.96	0.35	0.000	5384
Sex	0.50	0.48	-0.02	0.311	5684
SES	0.70	0.68	-0.02	0.008	5249

Attritors are individuals who did not attend the respective data collection wave. Non-attritors are individuals who attended. P-values come from regressions of each baseline characteristic on attrition-group indicators, including strata fixed effects. Standard errors are clustered at the teacher level for student-level regressions.

Table A3: Attritors vs non-attritors: Follow-up

Variable	Non-Attritors	Structurally Ineligible	Conventional Attritors	P-value Ineligible vs Non-Attritors	P-value Conventional vs Non-Attritors	N
Teacher tests						
Baseline score	0.62		0.54		0.950	338
Age	46.71		46.03		0.359	336
Sex	0.61		0.71		0.655	338
Classroom observations						
Baseline score	0.64	0.54	0.54	0.156	0.222	338
Age	46.93	45.92	46.03	0.289	0.216	336
Sex	0.58	0.63	0.78	0.708	0.014	338
Student tests						
Baseline score	0.30	0.28	0.27	0.201	0.001	6010
Age	10.60	10.60	11.09	0.628	0.000	5384
Sex	0.49	0.50	0.49	0.997	0.768	5684
SES	0.70	0.70	0.69	0.237	0.477	5249

Attritors are individuals who did not attend the respective data collection wave. Non-attritors are individuals who attended. At follow-up, attrition is split into structurally ineligible attrition and conventional attrition. Structurally ineligible attritors are individuals who could not be re-assessed because they no longer met the study eligibility criteria (e.g. teachers no longer teaching mathematics or students assigned to a different mathematics teacher). Conventional attritors are all other individuals who did not attend the assessment. No structurally ineligible attritors exist at endline or for teacher tests. P-values come from regressions of each baseline characteristic on attrition-group indicators, including strata fixed effects. Standard errors are clustered at the teacher level for student-level regressions.

Table A4: Attritor characteristics per experimental group: Endline

	Control	T1: Math	T2: Pedagogy	T3: Combined	P-value	N
Teacher tests						
Baseline score	0.53	0.64	0.50	0.65	0.21	29
Age	39.60	50.00	47.89	43.33	0.50	28
Sex	0.70	0.50	0.78	0.67	0.36	29
Classroom observations						
Baseline score	0.57	0.47	0.57	0.67	0.02	29
Age	39.88	45.40	48.50	39.40	0.73	29
Sex	0.50	1.00	0.50	0.70	0.59	29
Student tests						
Baseline score	0.24	0.27	0.25	0.25	0.95	1060
Age	11.20	10.76	10.99	10.99	0.34	2079
Sex	0.47	0.48	0.47	0.48	0.93	2234
SES	0.69	0.74	0.70	0.71	0.08	2069

The penultimate column indicates the p-value of a joint F-Test that each treatment dummy coefficient is equal to 0. Balance is calculated using strata fixed effects. Standard errors are clustered at the teacher level. At follow-up, the table includes only conventional attritors. Structurally ineligible attritors (e.g. teachers no longer teaching mathematics or students assigned to a different mathematics teacher) are excluded.

Table A5: Attritor characteristics per experimental group: Follow-up

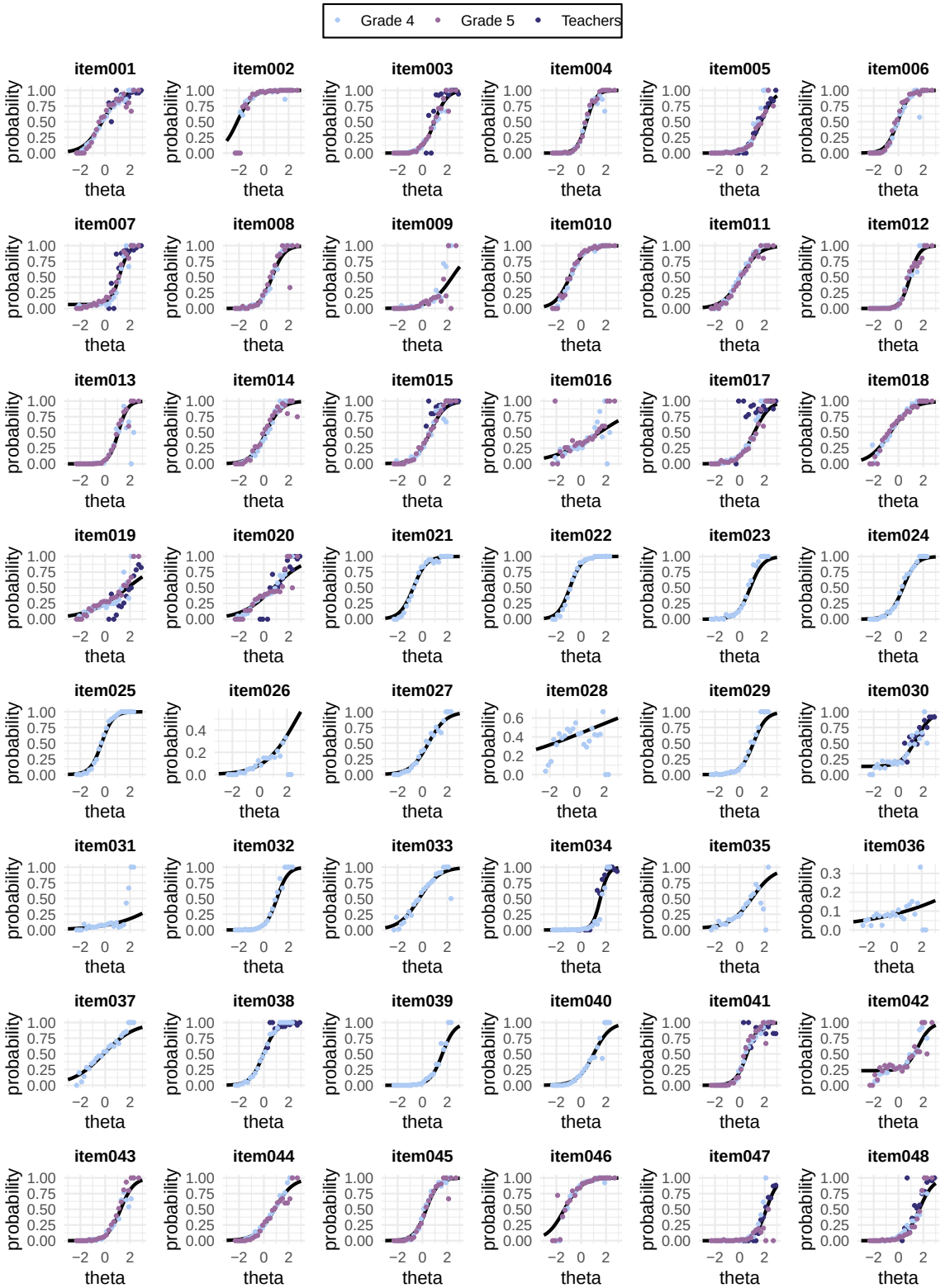
	Control	T1: Math	T2: Pedagogy	T3: Combined	P-value	N
Teacher tests						
Baseline score	0.52	0.61	0.55	0.51	1.00	41
Age	39.62	53.00	45.92	46.33	0.23	39
Sex	0.62	0.62	0.75	0.77	0.78	41
Classroom observations						
Baseline score	0.63	0.52	0.52	0.49	0.13	58
Age	37.21	49.56	50.27	46.31	0.01	58
Sex	0.71	0.81	0.73	0.85	0.72	58
Student tests						
Baseline score	0.24	0.29	0.25	0.30	0.17	847
Age	10.93	11.07	11.16	11.13	0.63	837
Sex	0.47	0.51	0.49	0.45	0.52	900
SES	0.68	0.70	0.71	0.68	0.85	809

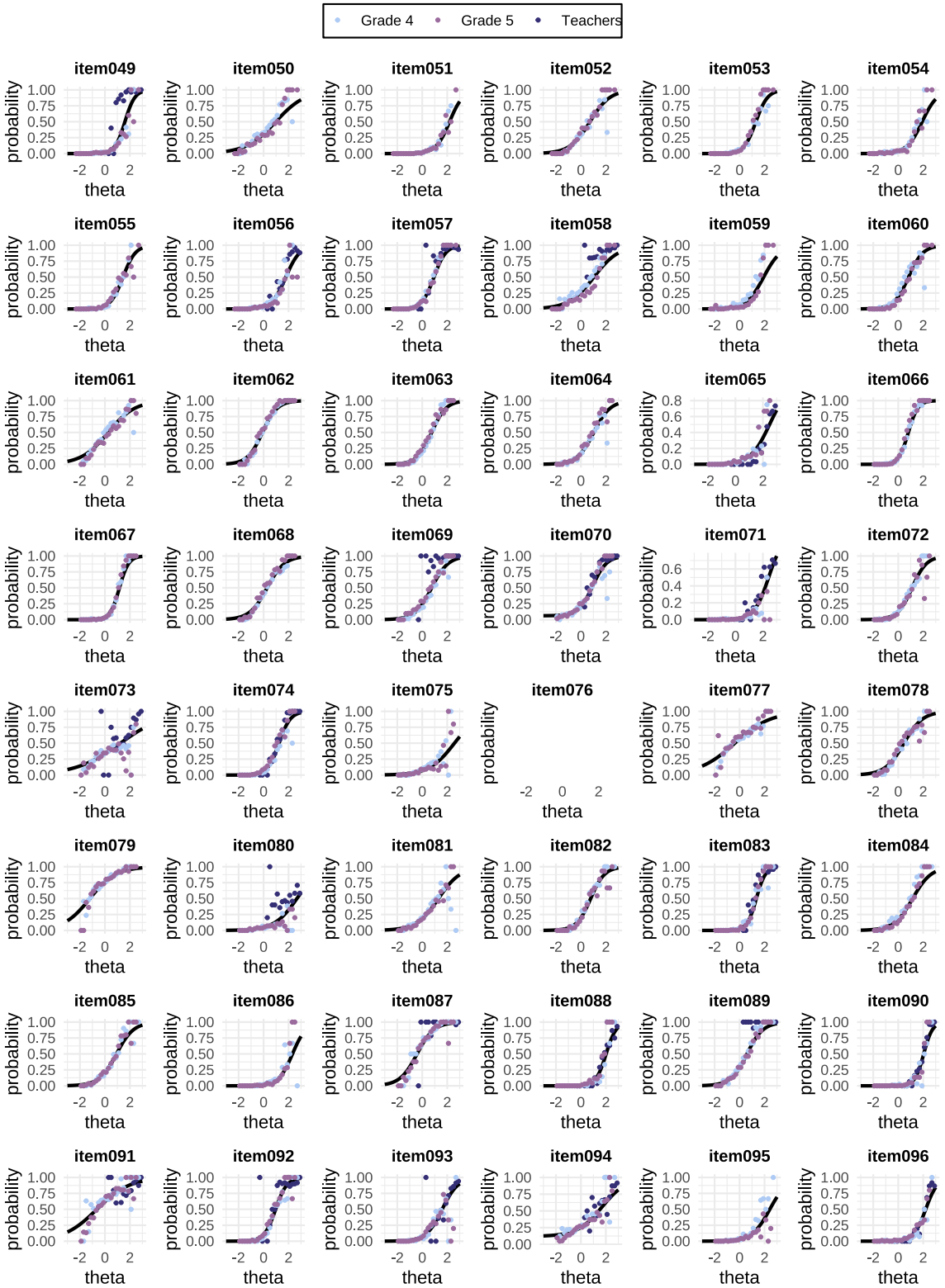
The penultimate column indicates the p-value of a joint F-Test that each treatment dummy coefficient is equal to 0. Balance is calculated using strata fixed effects. Standard errors are clustered at the teacher level. At follow-up, the table includes only conventional attritors. Structurally ineligible attritors (e.g. teachers no longer teaching mathematics or students assigned to a different mathematics teacher) are excluded.

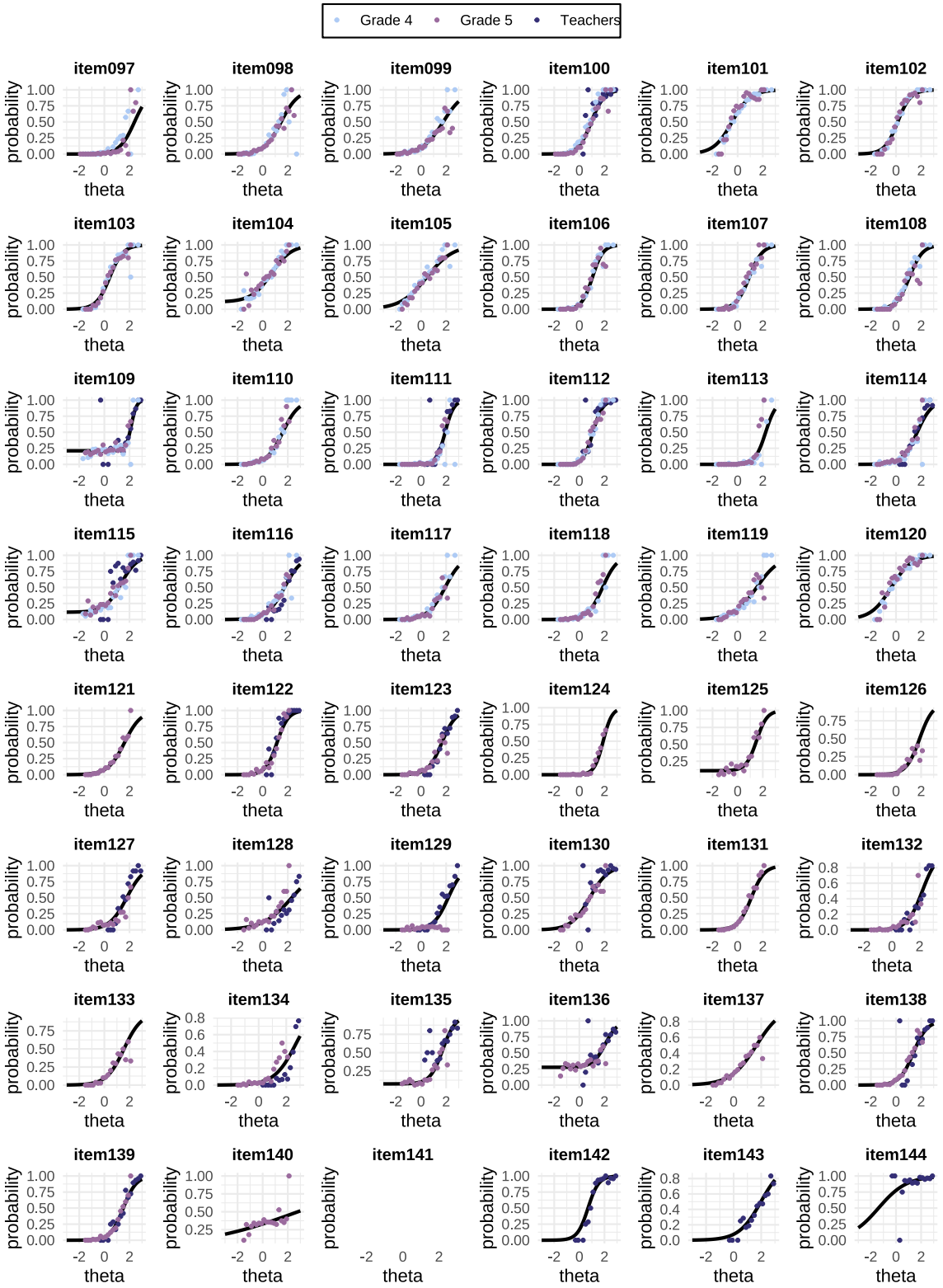
A5 Item Response Theory

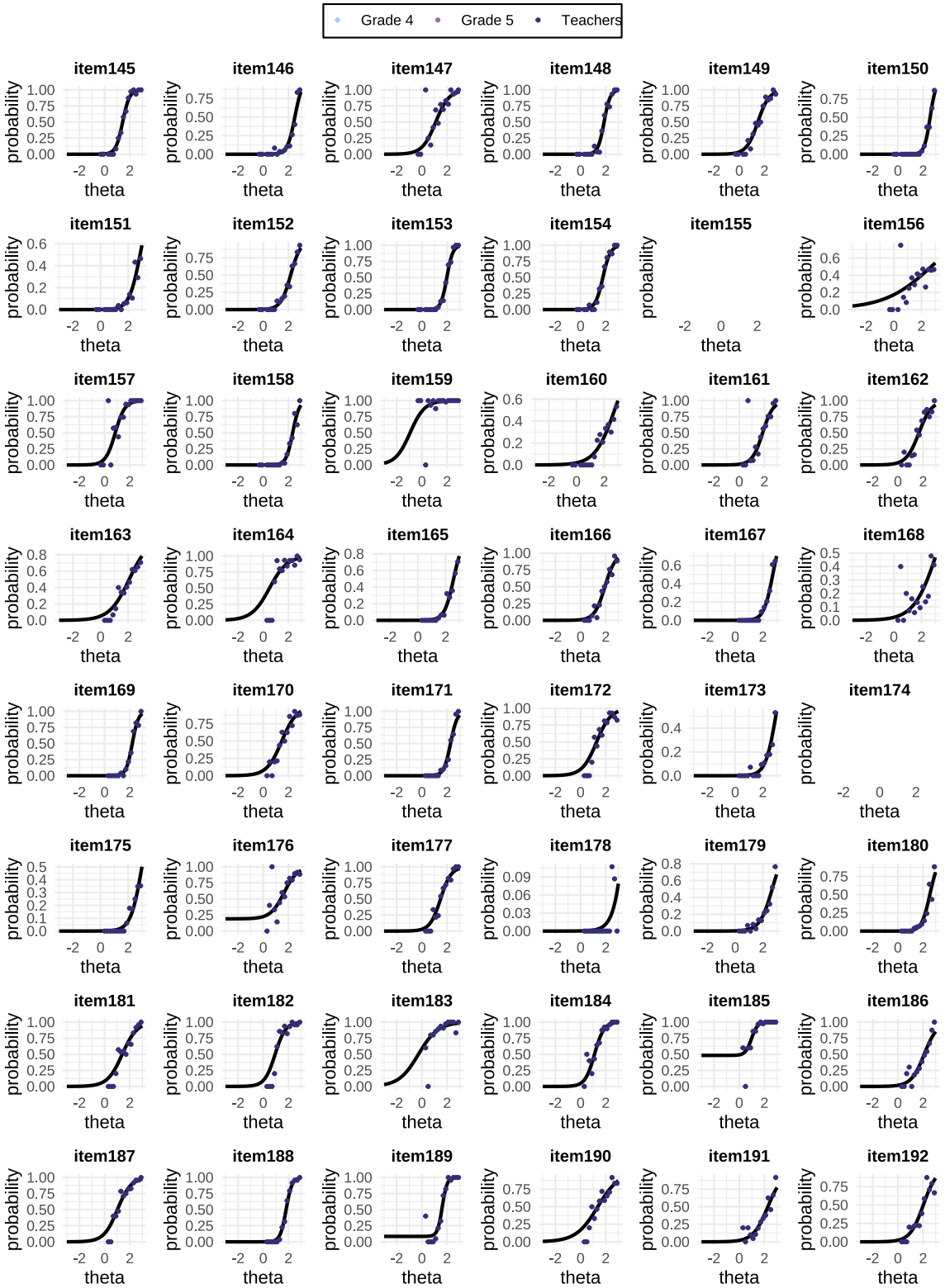
A5.1 IRT Linking

Figure A5: DIF plots




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Grade 4
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Grade 4
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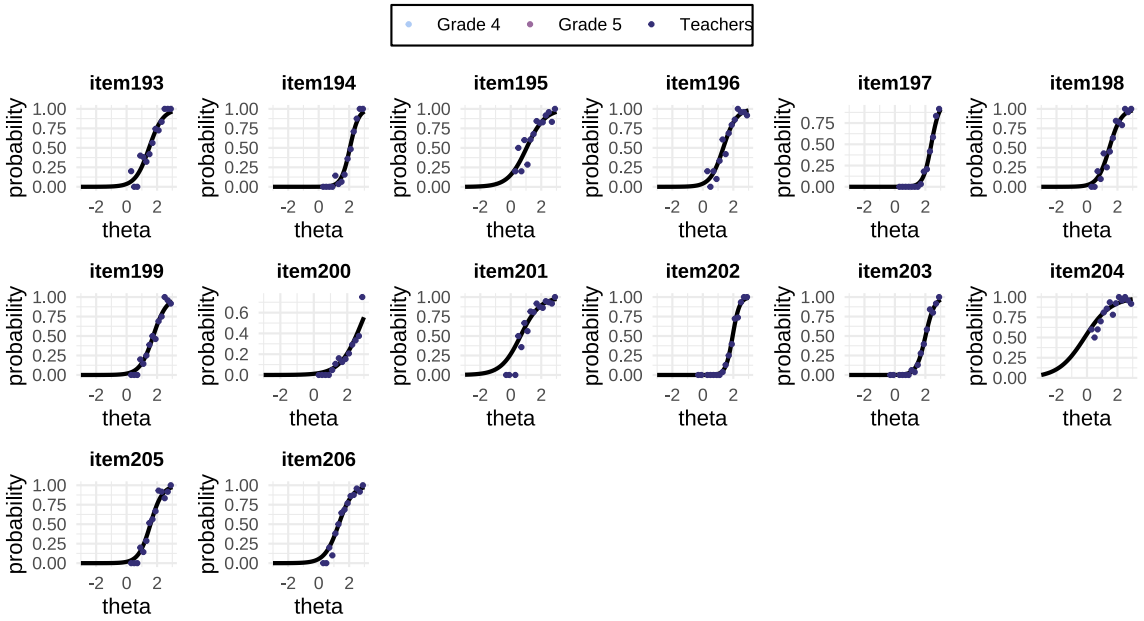


Figure A6: DIF plots

The figure compares empirical item characteristic curves by respondent group (Grade 4, Grade 5, and teachers) with the pooled IRT item characteristic curve. Close alignment indicates little evidence of differential item functioning (DIF).

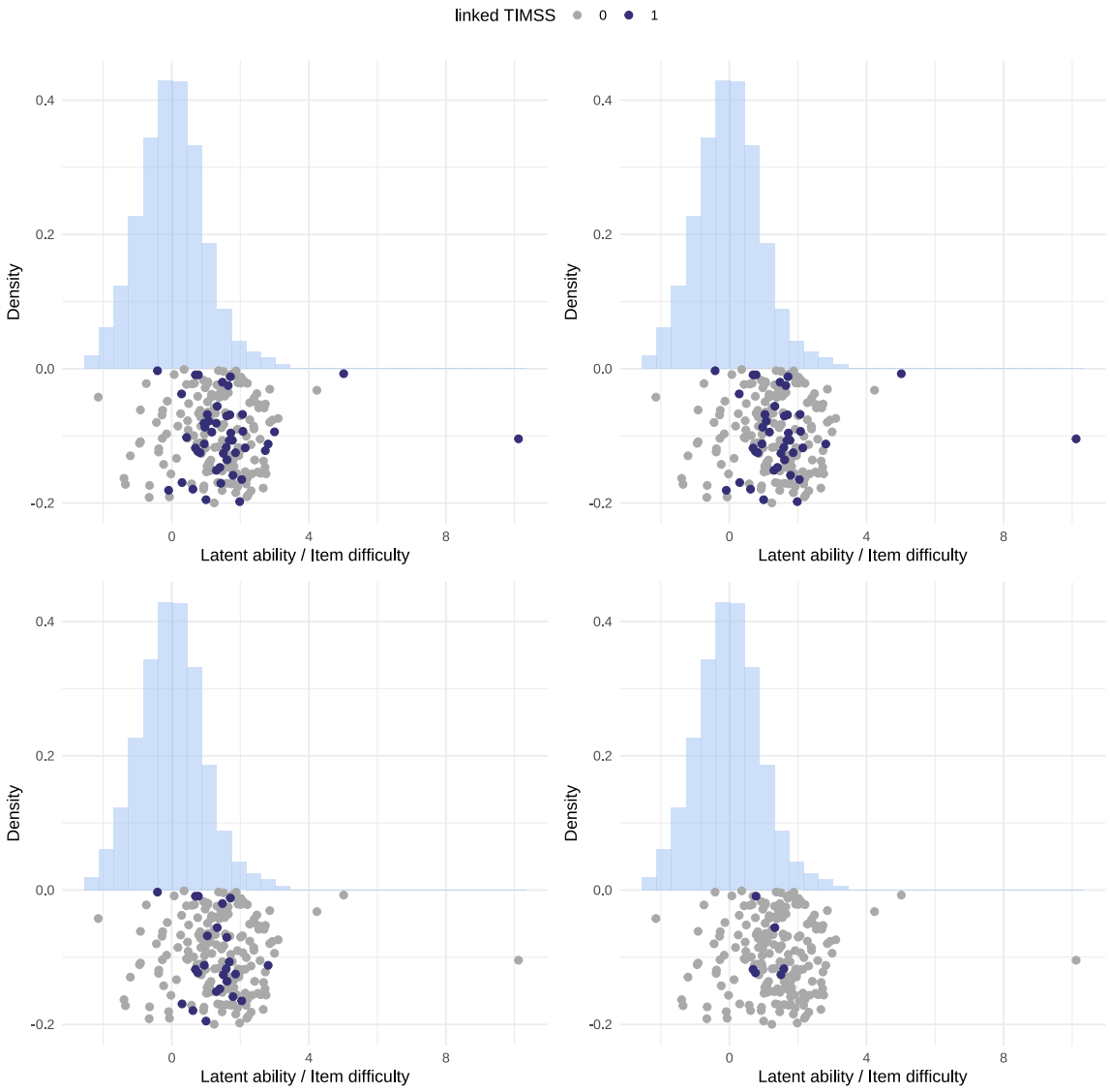


Figure A7: Wright map

The Wright Map, or person–item map, displays the distribution of the latent trait in the population (top part of the figure) alongside the distribution of item difficulties in the test (bottom part of the figure). The top left panel highlights all TIMSS items, while the top right restricts attention to TIMSS items without substantial DIF. The bottom left panel highlights only items with very little or no DIF, and the bottom right highlights those with no DIF. All difficulty parameters are taken from the freely estimated model in Section 2.4, which does not impose TIMSS linking.

A5.2 TIMSS Linking

Our model incorporates a 25% overlap with the 2015 Trends in International Mathematics and Science Study (TIMSS) items (Grønmo et al., 2015), enabling us to additionally project item scores onto the TIMSS scale using IRT. This allows us to situate

students' and teachers' performance in an international context and to express effect sizes on a common, externally benchmarked scale. While TIMSS primarily focuses on high-income countries, it has also been conducted in several low- and middle-income settings, such as Jordan, Morocco, and Indonesia.

To place our estimated abilities on the TIMSS scale, we adopt a common item nonequivalent groups design (Lee and Lee, 2018). In this design, two cohorts, which may differ in their underlying ability distributions, take two test forms: an old form O and a new form N. These forms include an overlapping set of anchor items. The old form is usually calibrated first, at a different point in time. The new form is then calibrated while holding the anchor item parameters fixed at their previously estimated values. In our context, the international TIMSS assessment 2015 serves as form O and our study exams serve as form N. Because the anchor items retain their parameters from the TIMSS calibration, all item and ability estimates for the new form are placed directly on the TIMSS scale. Kim (2006) and Lee and Lee (2018) emphasize that fixed calibration performs well only when the latent ability distribution is allowed to adjust during estimation, rather than being fixed at the default N(0,1) distribution used by many IRT software packages. When the distribution is held fixed, the model effectively forces the new sample to match a normal distribution, causing item parameters to shift in order to accommodate any mismatch, and placing new items on the wrong scale. Updating the distribution during estimation instead lets the data reveal where the new group sits relative to the anchor items, ensuring that new items are placed on the correct scale. Doing this correctly is particularly important in our setting, where students in El Salvador perform far below the TIMSS reference population.[18]

After this step, we perform a linear transformation to map the estimated IRT scores onto the actual TIMSS scale. Following the approach outlined by Patel and Sandefur (2020), we use plausible value averages from the 2015 TIMSS assessment, setting the mean as the intercept and the standard deviation as the slope parameter for the transformation.

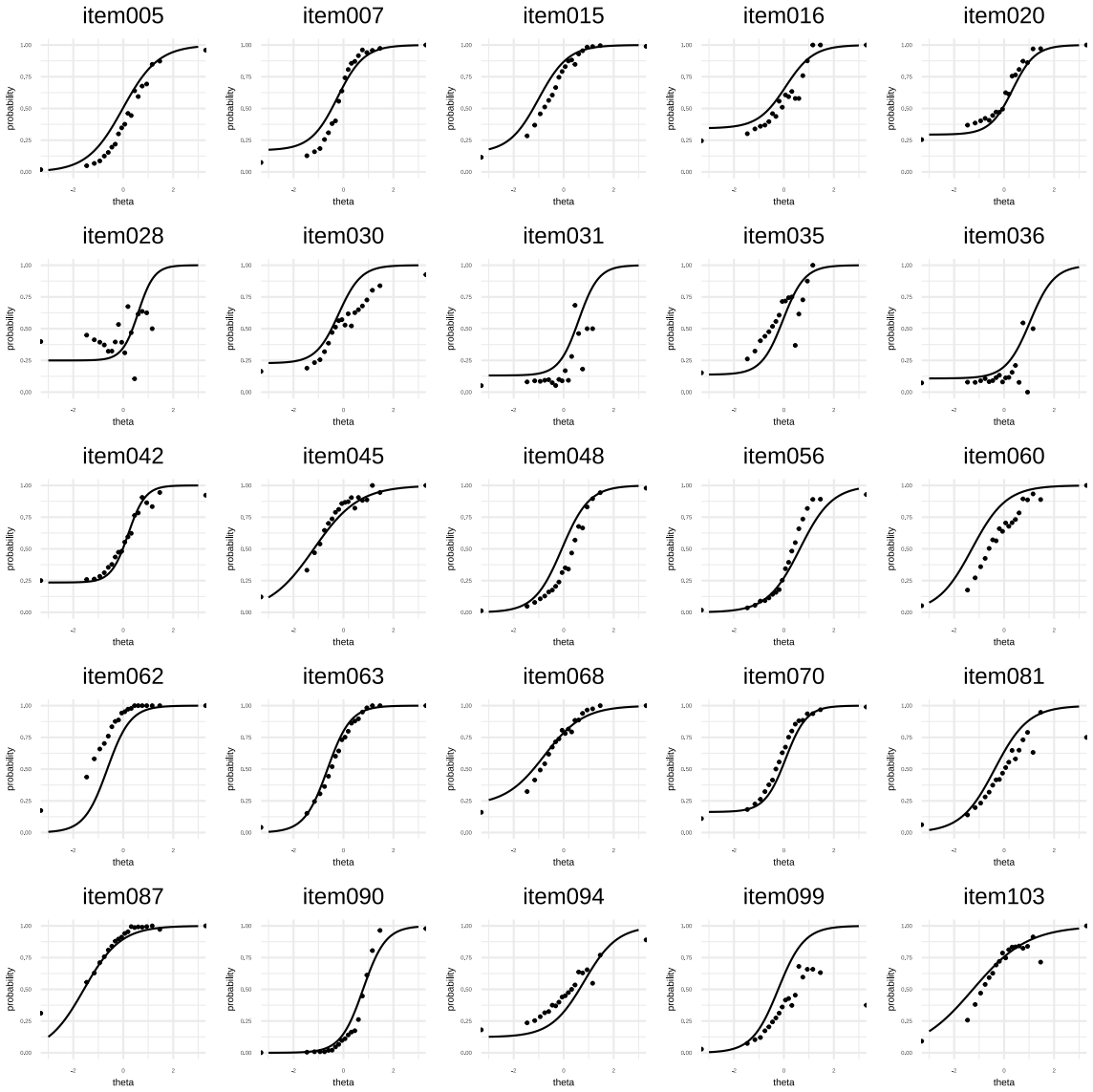
For our mappings to be valid, the anchor items must be population invariant. In practice, this requires that the items function equivalently and measure the same construct in both populations. An intuitive illustration is a math item with substantial text: if students in our Salvadorian sample have weak reading skills, the item begins

[18]Our approach departs from Patel and Sandefur (2020), who impose a fixed ability distribution of $N(0, 1)$. When we compare IRT scores under a fixed versus an updated distribution, the resulting estimates differ markedly (with average baseline scores after accounting for DIF of 457.5 versus 381.2, respectively) .

to load on a second construct, so that it no longer functions in the same way as in TIMSS. We assess population invariance by examining differential item functioning (DIF). Using the TIMSS parameter estimates, we plot the item characteristic curves for the anchor items and compare them with empirical curves for our sample. If the curves align, DIF is unlikely to be a concern. However, many of our anchor items display substantial DIF (see Figure A8), suggesting that the resulting estimates may be biased.[19] To address this, we categorize all anchor items into four groups based on the severity of DIF: substantial, moderate, very little, and none. We then estimate four alternative mappings, each using a different subset of items as linking items. One model uses the full set of TIMSS items, while the other three successively restrict the linking set to items in the highest-quality DIF categories. The last two categories (very little and none) contain only six and five items, which generates a familiar variance-bias trade-off: retaining fewer items reduces DIF but also yields a less stable link. The main divergence arises between the unrestricted specification and the restricted ones. When outcomes are grouped into four percentiles, the first specification agrees with the other three in only 62–78% of classifications depending on the model[20] , whereas the restricted specifications agree with one another in 84–92% of cases. We select the model that retains the best three item categories for linking, thereby excluding items with substantial DIF, which constitute more than half of the TIMSS pool. This leaves 19 linking items. The resulting set strikes a balance between limiting DIF and maintaining a sufficiently large and diverse group of items that span the difficulty range of our test (see Wright Map Figure A7), reflect the distribution of TIMSS item difficulties, and remain representative of TIMSS content. As in the free model in Section 2.4, the linking occurs primarily in the upper half of the distribution, which again implies that scores for lower-ability students are estimated with less precision.

[19]This is another key point of departure from Patel and Sandefur (2020). DIF plots are informative primarily for anchor items, where reference test ICCs can be compared with empirical curves under the fixed calibration. In the analysis of Patel and Sandefur (2020), only a small subset of plotted items are anchors, and these almost exclusively exhibit substantial DIF, consistent with our findings.

[20]The discrepancy is even larger when the latent ability distribution is fixed at $N(0, 1)$ rather than updated during estimation, with agreement falling to 35–53%.


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probability
theta

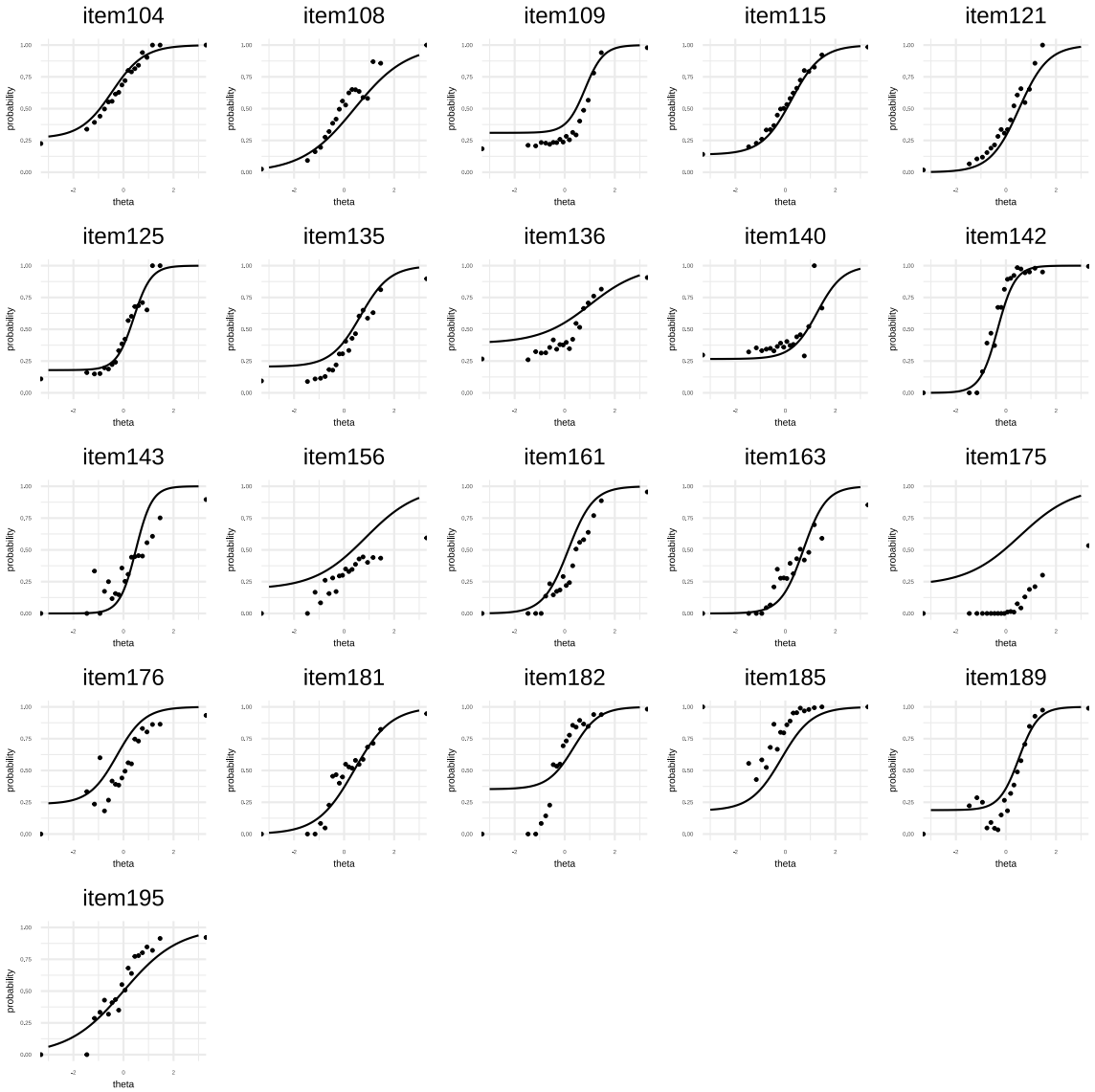


Figure A8: DIF plots for TIMSS linking (anchor items)

The figure compares empirical item characteristic curves for the TIMSS anchor items used in the linking procedure with the pooled IRT item characteristic curves. Close alignment indicates little evidence of differential item functioning (DIF).

A6 Survey and Classroom Observation Data

Table A6: Survey outcomes: Teachers

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Workload	0.415**	0.469***	0.540***	307	0.223	0.204	0.152	293
	(0.164)	(0.169)	(0.161)		(0.175)	(0.164)	(0.178)	
Teacher Impact Index	0.111	-0.140	0.072	308	-0.119	-0.297*	-0.053	296
	(0.166)	(0.159)	(0.162)		(0.176)	(0.172)	(0.168)	
Job Dissatisfaction Index	-0.195	0.008	0.155	305	-0.062	0.046	-0.077	293
	(0.140)	(0.158)	(0.156)		(0.148)	(0.169)	(0.161)	
Didactics Index	0.154	0.073	0.003	308	0.081	0.107	0.075	296
	(0.160)	(0.163)	(0.161)		(0.167)	(0.181)	(0.162)	
Preparation Index	0.087	0.022	-0.054	308	0.085	0.012	-0.068	296
	(0.150)	(0.167)	(0.158)		(0.151)	(0.148)	(0.153)	

All dependent variables are standardized unless stated otherwise. Survey indices are standardized summary indices constructed following Anderson (2008). The Teacher Impact Index combines teachers' perceived impact on students, mathematics learning, and didactics. The Job Dissatisfaction Index combines reverse-coded job satisfaction, frustration, perceived problems teaching mathematics, and perceived didactic problems. The Didactics Index combines reported use of activities, homework preparation, instructional materials, teaching methodologies, and examples. The Preparation Index combines use of ESMATE for preparation, preparation time, reverse-coded reports of preparing nothing, and lesson planning. Controls include strata fixed effects, baseline math score, and controls for teacher age and sex. Standard errors are clustered at the teacher level. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

Table A7: Survey outcomes: Teachers

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Workload	0.415**	0.469***	0.540***	307	0.223	0.204	0.152	293
	(0.164)	(0.169)	(0.161)		(0.175)	(0.164)	(0.178)	
Perceived impact on students								
Teacher	-0.008	-0.213	0.222	307	-0.170	-0.280*	-0.155	296
	(0.168)	(0.164)	(0.168)		(0.170)	(0.169)	(0.185)	
Didactics knowledge	0.086	-0.074	0.063	308	0.005	-0.163	0.129	296
	(0.176)	(0.165)	(0.153)		(0.172)	(0.184)	(0.162)	
Math knowledge	0.149	-0.021	-0.105	308	-0.122	-0.269	-0.096	296
	(0.162)	(0.156)	(0.159)		(0.173)	(0.182)	(0.166)	
Lesson preparation								
Time (in minutes)	-0.102	-0.015	0.024	301	0.157	-0.075	-0.052	277
	(0.152)	(0.160)	(0.155)		(0.177)	(0.154)	(0.196)	
Esmate usage	0.092	0.095	0.006	306	0.127	0.123	0.005	295
	(0.117)	(0.127)	(0.123)		(0.103)	(0.106)	(0.112)	
Own planning	0.059	-0.040	-0.215	300	-0.107	-0.042	-0.312*	290
	(0.157)	(0.162)	(0.178)		(0.161)	(0.184)	(0.169)	
Own activities	-0.038	0.019	-0.159	306	0.003	0.016	-0.050	292
	(0.156)	(0.150)	(0.151)		(0.160)	(0.174)	(0.156)	
Own didactic material	0.410**	0.214	0.101	306	0.281*	0.231	0.033	292
	(0.163)	(0.159)	(0.164)		(0.161)	(0.163)	(0.150)	
Homework check	0.090	-0.236	-0.031	304	-0.112	-0.149	-0.045	291
	(0.157)	(0.188)	(0.173)		(0.193)	(0.213)	(0.182)	
No Preparation	-0.096	0.040	-0.010	286	-0.028	-0.013	-0.168	271
	(0.153)	(0.177)	(0.164)		(0.161)	(0.164)	(0.160)	
During the lesson								
Satisfaction	-0.132	-0.411*	-0.333*	304	-0.135	-0.310*	0.014	293
	(0.207)	(0.215)	(0.189)		(0.158)	(0.186)	(0.163)	
Frustration	-0.197	0.022	0.132	303	0.032	0.068	-0.039	291
	(0.142)	(0.172)	(0.167)		(0.152)	(0.163)	(0.158)	
Methodology variation	0.072	0.216	0.006	303	0.054	0.218	0.172	294
	(0.159)	(0.153)	(0.160)		(0.173)	(0.166)	(0.172)	
Example usage	0.071	0.213	0.102	304	0.039	0.108	0.196	293
	(0.161)	(0.172)	(0.161)		(0.161)	(0.158)	(0.150)	
Difficulity didactics knowledge	-0.303**	-0.195	-0.073	304	-0.183	-0.200	-0.054	291
	(0.138)	(0.152)	(0.147)		(0.148)	(0.147)	(0.155)	
Difficultiy math knowledge	-0.116	-0.104	0.098	305	-0.108	-0.042	-0.223	291
	(0.142)	(0.154)	(0.149)		(0.153)	(0.166)	(0.149)	
Comparison to other subjects								
Time (in minutes)	-4.918	-5.735	-2.427	264	5.416	-2.834	3.877	251
	(6.590)	(6.743)	(7.096)		(7.698)	(6.084)	(12.210)	
Freedom	0.009	-0.139	0.104	264	-0.212	-0.040	0.089	262
	(0.170)	(0.163)	(0.178)		(0.172)	(0.162)	(0.177)	
Student interest	-0.065	0.172	0.267*	266	-0.171	-0.205	-0.293	265
	(0.168)	(0.162)	(0.152)		(0.164)	(0.159)	(0.181)	
Motivation	-0.165	-0.154	0.094	266	-0.118	-0.414**	-0.080	262
	(0.177)	(0.178)	(0.166)		(0.167)	(0.188)	(0.188)	
Methodologies	0.259*	0.006	0.296*	265	-0.006	-0.151	0.049	264
	(0.153)	(0.163)	(0.155)		(0.176)	(0.174)	(0.176)	
Class management	-0.062	-0.192	0.304*	266	-0.052	-0.274	-0.064	265
	(0.176)	(0.171)	(0.166)		(0.182)	(0.179)	(0.176)	

All the dependent variables are standardized unless stated otherwise. Controls include strata fixed effects, baseline math score and controls for teacher age and sex. Standard errors are clustered at the teacher level. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

Table A8: Survey outcomes: Students

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Favorite subject is math	0.034	0.010	-0.021	4321	-0.026	-0.104**	-0.062	2109
	(0.026)	(0.027)	(0.026)		(0.048)	(0.046)	(0.047)	
Least favorite subject is math	-0.033	0.015	0.014	3923	0.043	0.088*	0.056	1939
	(0.027)	(0.025)	(0.026)		(0.041)	(0.046)	(0.042)	
Favorite activity is school	0.024	-0.036	0.000	4360	0.043	0.010	0.000	2165
	(0.024)	(0.028)	(0.025)		(0.033)	(0.033)	(0.041)	
Least favorite activity is school	-0.009	0.019	0.018	4032	-0.018	0.006	-0.018	1976
	(0.016)	(0.018)	(0.017)		(0.024)	(0.025)	(0.027)	

Controls include strata fixed effects, the baseline value of the outcome, student sex, age, baseline math score and teacher baseline score. Missing values for age and sex are replaced with the control group mean, with missing-value indicators included in the regressions. Standard errors are clustered at the teacher level. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

Table A9: Classroom observations extended: Teachers

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	Control mean	T1: Math	T2: Ped	T3: Both	Control mean
Teacher use of time								
Active (%)	-0.011	0.002	0.015	0.452	-0.014	-0.059	0.020	0.528
	(0.036)	(0.036)	(0.036)		(0.038)	(0.039)	(0.039)	
Passive (%)	0.006	0.031	-0.023	0.360	0.026	0.048	0.016	0.207
	(0.035)	(0.036)	(0.036)		(0.032)	(0.033)	(0.033)	
Management (%)	0.011	-0.020	0.014	0.132	0.036	0.047	-0.001	0.143
	(0.025)	(0.025)	(0.025)		(0.028)	(0.029)	(0.029)	
Social (%)	-0.006	-0.013	-0.006	0.055	-0.028	-0.029	-0.024	0.084
	(0.015)	(0.015)	(0.015)		(0.018)	(0.019)	(0.019)	
Material use								
Interactive Material (%)	0.039	0.053**	0.052**	0.038	0.030	0.028	0.032	0.043
	(0.024)	(0.024)	(0.024)		(0.025)	(0.026)	(0.025)	
Traditional Material (%)	-0.056	-0.077**	-0.088**	0.748	-0.057	-0.052	-0.025	0.653
	(0.039)	(0.039)	(0.039)		(0.040)	(0.041)	(0.041)	
Material variation (sd)	0.087	0.177	0.037	0.000	0.009	-0.135	0.085	0.000
	(0.147)	(0.149)	(0.150)		(0.170)	(0.175)	(0.174)	
Pedagogy								
Does lesson outline (sd)	0.119	0.211	0.236	0.000	-0.009	0.145	-0.093	0.000
	(0.160)	(0.162)	(0.163)		(0.156)	(0.162)	(0.160)	
Does homework review (sd)	-0.146	-0.015	-0.002	-0.000	0.055	0.015	-0.008	0.000
	(0.165)	(0.167)	(0.168)		(0.172)	(0.178)	(0.176)	
Uses examples (sd)	0.204	0.202	0.230	0.000	-0.291	-0.098	-0.032	-0.000
	(0.154)	(0.156)	(0.157)		(0.189)	(0.194)	(0.194)	
Moves in classroom (sd)	0.023	0.130	0.147	0.000	0.061	0.227	0.266	0.000
	(0.153)	(0.154)	(0.155)		(0.166)	(0.171)	(0.170)	
Gives homework (sd)	0.114	0.317**	0.231	0.000	-0.181	0.123	-0.271	0.000
	(0.148)	(0.150)	(0.151)		(0.169)	(0.173)	(0.172)	
Does lesson summary (sd)	0.128	0.295*	0.267	0.000	-0.186	-0.061	-0.162	0.000
	(0.160)	(0.162)	(0.164)		(0.178)	(0.182)	(0.182)	
Activities								
Activity variation (sd)	0.178	-0.024	0.061	0.000	0.177	-0.056	0.014	-0.000
	(0.142)	(0.143)	(0.144)		(0.159)	(0.163)	(0.163)	
Conducts interactive activity	0.007	0.182***	0.118**	0.092	0.096*	0.145***	0.054	0.018
	(0.058)	(0.059)	(0.059)		(0.050)	(0.051)	(0.051)	
Distracted students								
One or more students distracted (%)	-0.008	-0.077*	-0.045	0.447	0.008	-0.055	-0.044	0.707
	(0.046)	(0.047)	(0.047)		(0.046)	(0.047)	(0.047)	
Group distracted (%)	-0.012	-0.067	-0.050	0.368	0.044	-0.019	-0.028	0.592
	(0.044)	(0.045)	(0.045)		(0.053)	(0.054)	(0.054)	

Sample sizes are n = 307 for the estimation of endline effects and n = 230 for follow-up effects. All the dependent variables are either standardized (sd), binary or indicated as a percentage of the lesson time (%). Interactive material constitutes shared material, learning aids or ICT. Traditional material consitutes textbook, notebook and blackboard. Controls include strata fixed effects and demographic teacher variables.

A7 Main Effects

Table A10: ITT estimates: Results per grade

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Math: Share correct responses (\%)								
4th grade	0.166*	0.101	-0.034	2443	0.004	0.007	-0.112	1188
	(0.085)	(0.096)	(0.081)		(0.112)	(0.122)	(0.142)	
5th grade	-0.137*	-0.123	-0.168**	2507	-0.013	-0.007	-0.071	1244
	(0.083)	(0.095)	(0.085)		(0.108)	(0.096)	(0.118)	
Math: IRT score (sd)								
4th grade	0.135*	0.038	-0.026	2443	0.003	-0.054	-0.159	1188
	(0.081)	(0.094)	(0.083)		(0.105)	(0.111)	(0.136)	
5th grade	-0.149**	-0.149*	-0.176**	2507	0.005	-0.012	-0.015	1244
	(0.076)	(0.088)	(0.082)		(0.102)	(0.101)	(0.112)	
Math: TIMSS score (points)								
4th grade	9.805	-1.660	-1.106	2443	1.822	-5.737	-16.219	1188
	(7.026)	(8.338)	(7.204)		(8.849)	(9.778)	(11.714)	
5th grade	-10.560	-12.751*	-14.309*	2507	1.822	-5.737	-16.219	1188
	(6.674)	(7.501)	(7.409)		(8.849)	(9.778)	(11.714)	

Controls include strata fixed effects and the outcome variable at baseline. Standard errors are clustered at the teacher level. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

Table A16: Teacher content knowledge and student learning gains: Evidence from quasi-experimental data

Study	Student Effect (+ 1σ teacher score)	Country/Region	Grade	Empirical Strategy
Metzler and Woessmann (2012)	Math: 0.09, Language: 0.03 (insig.)	Peru	Grade 6	Teacher FE + Student FE
Bietenbeck, Piopiunik, and Wiederhold (2018)	Mixed: 0.03	6 East African countries	Grade 6	Teacher FE + Student FE
Bold et al. (2019)	Mixed: 0.07	7 African countries	Grade 4	Teacher FE + Student FE
Bau and Das (2020)	Math: 0.09, Language: 0.06	Pakistan, Punjab	Grade 3-5	Teacher value-added approach
Brunetti et. al (2023)	Math: 0.09–0.12	El Salvador, Morazán	Grade 3- 6	Various controls
Our results	Math: 0..07-0.08	El Salvador, Morazán	Grade 4-6	Various controls

Table A11: ITT estimates pedagogical exam: Disagregated

	Follow-up			
	T1: Math	T2: Ped	T3: Both	N
Lesson Plan (Part 1)				
Exam completness	0.291*	0.481***	0.291*	297
	(0.148)	(0.172)	(0.170)	
Overall quality	0.420***	0.420**	0.379**	297
	(0.156)	(0.166)	(0.170)	
Pedagogy score	0.616***	0.561***	0.580***	297
	(0.188)	(0.197)	(0.189)	
Structure	0.244*	0.442***	0.285*	297
	(0.142)	(0.155)	(0.163)	
Practical planning	0.347**	0.559***	0.213	297
	(0.169)	(0.191)	(0.180)	
Classroom Management (Part 2)				
Number of suggested measures	0.103	0.246	0.086	297
	(0.160)	(0.178)	(0.169)	
Measure apropriateness	0.319*	0.297	0.321*	295
	(0.176)	(0.192)	(0.169)	

Controls include strata fixed effects, the outcome variable at baseline and demographic controls. Standard errors are clustered at the teacher level. The overall quality evaluates the clarity and effectiveness of the lesson, including introduction, content, practice, and summary. The pedagogy score refers to the use of diverse methods, student differentiation, engagement, and material variety. Structure refers to how well the lesson plan is structured. Practical planning measures definition of objectives, time allocation, and material listing. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

Table A12: 2SLS estimates: Student math scores

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Without Controls								
Math: Share correct responses (%)	0.033	-0.002	-0.145	4950	0.006	-0.009	-0.044	2441
	(0.077)	(0.098)	(0.091)		(0.095)	(0.111)	(0.127)	
Math: IRT score (sd)	0.010	-0.059	-0.140	4950	0.018	-0.051	-0.033	2441
	(0.073)	(0.093)	(0.090)		(0.093)	(0.113)	(0.123)	
Math: TIMSS score (points)	1.021	-8.300	-11.795	4950	2.319	-6.655	-2.821	2441
	(6.516)	(8.271)	(8.143)		(8.535)	(10.555)	(11.292)	
With Controls								
Math: Share correct responses (%)	0.033	-0.006	-0.145	4950	0.013	0.002	-0.032	2441
	(0.077)	(0.098)	(0.091)		(0.093)	(0.109)	(0.125)	
Math: IRT score (sd)	0.011	-0.063	-0.140	4950	0.026	-0.040	-0.021	2441
	(0.072)	(0.093)	(0.090)		(0.091)	(0.111)	(0.122)	
Math: TIMSS score (points)	1.095	-8.735	-11.834	4950	2.960	-5.604	-1.613	2441
	(6.497)	(8.265)	(8.177)		(8.384)	(10.412)	(11.173)	

Endogenous treatment variables indicate assignment to the respective treatment arm interacted with an indicator for attending at least four of seven workshops. These endogenous variables are instrumented with the original treatment assignment indicators. Controls include strata fixed effects, the outcome variable at baseline, and demographic controls. Standard errors are clustered at the teacher level. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

Table A13: ITT estimates: Student math scores at endline, full sample and follow-up sample

	Endline				*Endline: Follow-up Sample*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Without Controls								
Math: Share correct responses (%)	0.028	0.001	-0.097	4950	0.058	-0.017	-0.066	2150
	(0.065)	(0.076)	(0.062)		(0.084)	(0.114)	(0.100)	
Math: IRT score (sd)	0.009	-0.043	-0.094	4950	0.053	-0.048	-0.035	2150
	(0.062)	(0.072)	(0.061)		(0.077)	(0.103)	(0.093)	
Math: TIMSS score (points)	0.910	-6.208	-7.960	4950	5.154	-5.958	-1.948	2150
	(5.535)	(6.392)	(5.576)		(6.907)	(9.083)	(7.986)	
With Controls								
Math: Share correct responses (%)	0.029	-0.002	-0.098	4950	0.059	-0.015	-0.059	2150
	(0.065)	(0.076)	(0.062)		(0.084)	(0.113)	(0.101)	
Math: IRT score (sd)	0.010	-0.046	-0.095	4950	0.055	-0.046	-0.027	2150
	(0.061)	(0.072)	(0.061)		(0.077)	(0.103)	(0.094)	
Math: TIMSS score (points)	0.974	-6.542	-7.990	4950	5.325	-5.777	-1.076	2150
	(5.519)	(6.381)	(5.600)		(6.912)	(9.070)	(8.070)	

The restricted sample includes only students who attended the follow-up exam. Controls include strata fixed effects, the outcome variable at baseline, sex and age fixed effects, and the teacher baseline score. Standard errors are clustered at the teacher level. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

Table A14: ITT estimates: Teachers subscores

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Number Sense & Place Value	0.102	-0.122	0.180	308	-0.072	0.028	-0.190	297
	(0.118)	(0.109)	(0.121)		(0.135)	(0.138)	(0.137)	
Addition & Subtraction	0.029	0.035	0.022	308	0.059	-0.014	0.054	297
	(0.150)	(0.162)	(0.152)		(0.144)	(0.154)	(0.146)	
Multiplication & Division	-0.032	-0.199	0.010	308	0.326**	-0.203	0.155	297
	(0.129)	(0.135)	(0.136)		(0.138)	(0.155)	(0.155)	
Fractions & Decimals	0.123	-0.141	0.092	308	0.273***	0.068	0.182**	297
	(0.081)	(0.086)	(0.084)		(0.084)	(0.099)	(0.083)	
Algebraic Thinking	0.110	-0.053	-0.109	308	0.173	0.123	0.085	297
	(0.120)	(0.111)	(0.114)		(0.122)	(0.117)	(0.122)	
Order of Operations / Combined Operations	-0.149	-0.040	-0.109	308	-	-	-	-
	(0.149)	(0.161)	(0.155)					
Geometry	0.218**	-0.046	0.253**	308	0.275**	0.137	0.179	297
	(0.109)	(0.108)	(0.106)		(0.110)	(0.109)	(0.113)	
Measurement & Units	0.271**	0.126	0.156	308	0.118	-0.136	0.162	297
	(0.132)	(0.147)	(0.141)		(0.165)	(0.169)	(0.173)	
Data & Statistics	0.076	-0.124	0.003	308	0.130	-0.171	-0.083	297
	(0.110)	(0.119)	(0.117)		(0.144)	(0.128)	(0.137)	
Proportionality & Percentage	0.083	0.024	-0.059	308	0.186	0.137	0.039	297
	(0.115)	(0.117)	(0.105)		(0.131)	(0.136)	(0.131)	

All the dependent variables are standardized unless stated otherwise. Controls include strata fixed effects, teacher sex and age and baseline math score. Standard errors are clustered at the teacher level.$*p < 0.1$; $**p < 0.05$; $***p < 0.01$

Table A15: ITT estimates: Students subscores

	Endline				*Follow-up*			
	T1: Math	T2: Ped	T3: Both	N	T1: Math	T2: Ped	T3: Both	N
Number Sense & Place Value	0.034	-0.023	-0.072	4950	0.035	-0.005	0.000	2441
	(0.059)	(0.060)	(0.059)		(0.082)	(0.082)	(0.088)	
Addition & Subtraction	0.052	0.004	-0.096*	4950	0.044	-0.081	-0.090	2441
	(0.060)	(0.063)	(0.058)		(0.075)	(0.078)	(0.090)	
Multiplication & Division	0.004	0.033	-0.054	4950	-0.034	-0.090	-0.091	2441
	(0.048)	(0.056)	(0.047)		(0.081)	(0.078)	(0.084)	
Fractions & Decimals	0.037	0.066	-0.091	4950	0.035	-0.049	-0.081	2441
	(0.063)	(0.075)	(0.061)		(0.080)	(0.077)	(0.096)	
Algebraic Thinking	-0.005	-0.054	-0.084	4950	-0.010	-0.007	-0.016	2441
	(0.065)	(0.064)	(0.058)		(0.061)	(0.058)	(0.069)	
Order of Operations / Combined Operations	0.087	-0.035	-0.104*	4950	0.117	0.207	0.058	1197
	(0.077)	(0.071)	(0.063)		(0.129)	(0.186)	(0.127)	
Geometry	0.006	0.028	-0.004	4950	0.032	0.096	0.111	2441
	(0.069)	(0.078)	(0.067)		(0.086)	(0.083)	(0.110)	
Measurement & Units	0.003	0.031	0.057	4950	-0.010	0.100	0.107	2441
	(0.050)	(0.051)	(0.055)		(0.084)	(0.115)	(0.093)	
Data & Statistics	0.041	0.026	0.026	4950	0.002	0.089	0.051	2441
	(0.046)	(0.057)	(0.050)		(0.062)	(0.068)	(0.076)	
Proportionality & Percentage	-	-	-	-	-0.119	-0.004	-0.046	1244
					(0.083)	(0.089)	(0.109)	

All the dependent variables are standardized. Controls include strata fixed effects, the baseline math score and teacher baseline math score. Standard errors are clustered at the teacher level. In the follow-up exams, Data & Statistics items only appeared in grade 5 exams, so data is only available for the younger cohort. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

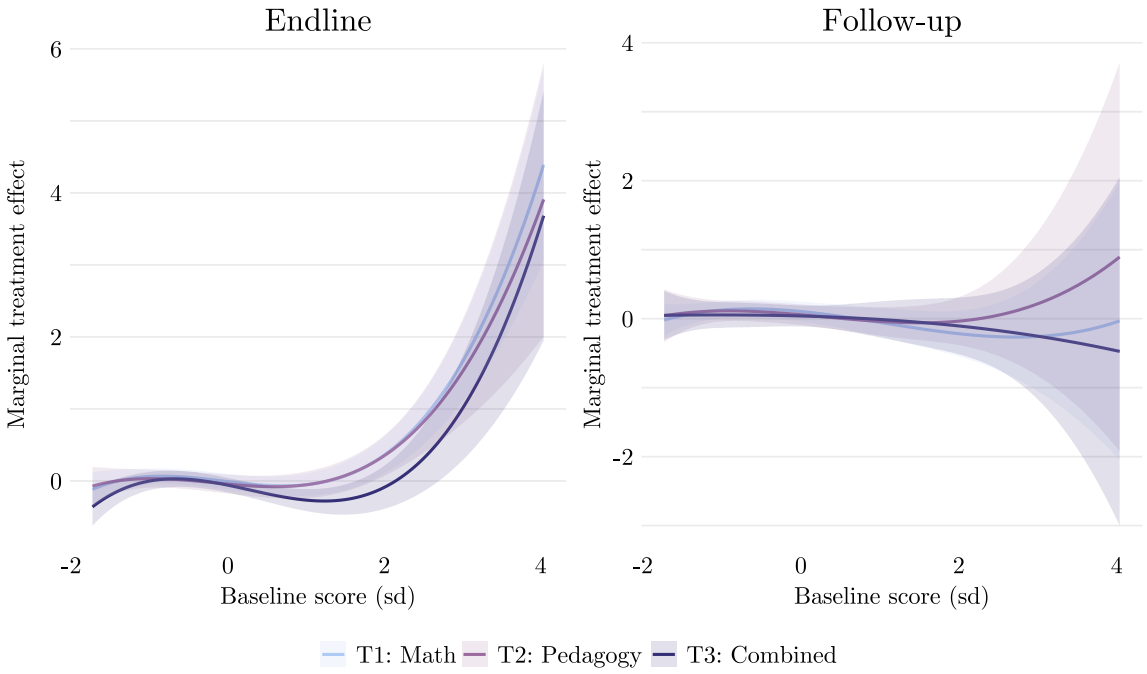


Figure A9: Endline effect heterogeneity over baseline student performance

Notes: The figure plots marginal treatment effects estimated from regressions interacting treatment indicators with a third-order polynomial in students' standardized baseline scores (percent correct); shaded areas denote 90% confidence intervals.

A8 Semi-Structured Interviews with Teachers

	General impression	Exemplary quotes on opportunities / likes	Exemplary quotes on challenges / dislikes
T1: Math	*«It has been very important for me because I have acquired new knowledge, and this is such a good tool that they have given us first and second cycle teachers that has helped us a lot in the classroom. » (Juana)*	*« […] they were all very interesting topics, topics where we, perhaps because we are not specialists in the area, were impressed to see the contents and learn a little more in each one of them. Each day was very useful and the topics that the teacher developed each one of them was really interesting. » (Norma)*	No negative aspects mentioned.
T2: Pedagogy	*«For me the impression is that it has been a well-organized, well-structured process. From the first session we felt the excitement of being able to attend the following sessions, it was very interesting, and it has awakened our curiosity to be able to continue implementing some knowledge that we already had but being able to convert it into something more practical with our students. » (Jorge)*	*«I liked everything that we were taught, all the didactic methodologies implemented by the teacher and the participation of all the classmates. » (Maria Teresa)*	No negative aspects mentioned.
T3: Both	*«I've never received training in this form that is so complete, because in addition to the virtual counselling, there are the face-to-face […] classes and the mathematical and didactic techniques and strategies as well. » (Patricia)*	*«I quite liked the fact that every day that the classes were given, they always applied concrete material. We reinforced it through, let's say, playing, through moving things. […] One of these activities was the use of the geoboard, which I found quite interesting. » (Ana)*	No negative aspects mentioned.

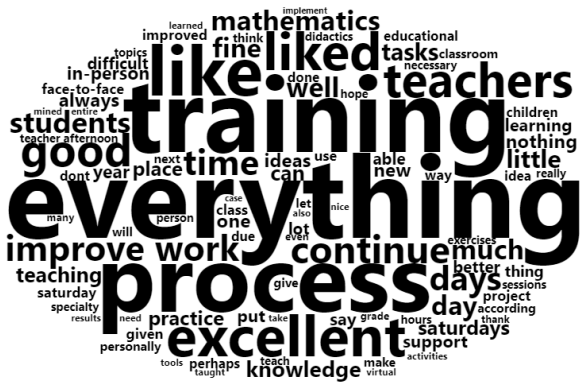
Figure A10: General questions

	Impact on didactical and / or mathematical knowledge	Impact on teaching	Impact on pupils	Training's impact relative to other basic education trainings
T1: Math	*«The mathematical knowledge was very nice, very broad. The trainings helped us to obtain new strategies, new techniques. […] the training was quite playful so that we could acquire more strategic and technical knowledge for the students. » (Ingrid)*	*« […] I have always been very persistent in making my students learn mathematics, but I didn't have the tool that would work more effectively and with this training, it has strengthened me more and I have managed to get the students to do what I wanted them to do: to like the subject. » (Juana)*	*«In this formative process with the students, they gave more importance to the subject of mathematics, and they saw it more as having it closer to them. They learnt that mathematics is beautiful, that you don't have to be afraid of it. » (Ingrid)*	*«I see it as something that helps us mathematics teachers, as we perhaps lack constant training, new strategies, and techniques to implement. » (Ingrid)*
T2: Pedagogy	*«In this process, we have been awakened to many strategies. […] [T]hey have given us the necessary tools to be able to take them to a different level; to be able to perform better in our educational work, in our schools […] where we work every day. » (Jorge)*	*« The most important thing, the learning pace of each student. Oftentimes we say that we do not achieve the objectives; that is where they tell us that we must look for new strategies for those [weak] students […] because not everyone learns in the same way. » (Jesús)*	*« They really liked the games and everything, they are more motivated. […] All this has helped us to motivate the children more and as they say, playing is also a way to learn, right? […] it has been noticed that there is more learning. » (Jesús)*	*« We felt that that we have learned more, because of the confidence they gave us to always ask questions, the doubts we have and more than anything the tasks were well based on what we are teaching. » (Jesús)*
T3: Both	*« My speciality is not mathematics, but this process taught me a lot about mathematics […]. For me that class was like something significant, and I won't forget how to work with fractions. » (Ruth)*	*[T]he students are no longer just in the classroom with their notebooks and the blackboard, but we also go outside the classroom […]. For example […] to form circles, so that they can expand their knowledge and assimilate what a perimeter is. That's already changing the old traditional teaching, isn't it? (Patricia)*	*«They liked it a lot, because regarding the multiplication tables game, they all wanted to go on to say the multiplication tables with the Buster game. Even the second graders felt more motivated and participated more. So, this has really changed the way we teach and motivate them with these dynamics. » (Patricia)*	*« I liked that they left guides that came with many exercises that can be implemented. I took some of them up again. I feel that this made quite a difference […] [I]t has been face-to-face, something that I have valued a lot, because virtually, you lose important aspects like […] friendships or bonds […] with other teachers […]. » (Johana)*

Figure A11: Questions on program impact

	Feedback on the seven days of face-to-face training	Feedback on the audiovisual material provided	Feedback on activities between sessions	Feedback on the tutoring provided
T1: Math	*«The seven days were excellent, and the explanations were very precise, and I believe that we all understood and reinforced the knowledge that they imparted to us. The teacher […] answered our questions, and provided us with all the information we needed. » (Juana)*	*«Very applicable, practical both for us as teachers to keep practising and improving our maths skills and to be able to teach and try to make it more useful for students' understanding and teaching. » (Julissa)*	*«Some of them were a bit complicated. We needed help from materials, from the notes we made and from some of our colleagues who were always there. […] But the truth is that everything worked well. » (Norma)*	*« As the tutoring sessions progressed, we realized what an excellent teacher he was. […] Whenever we called him with questions, he was always available and never left us on our own. » (Norma)*
T2: Pedagogy	*« Excellent, very dynamic, well-structured, everything is detailed, and the booklet helped us to be able to locate ourselves within each day. » (Jorge)*	*« Very special, because everything that was used was well-related to what we were learning, everything was very appropriate, and everything was there. » (Jesús)*	*« I think they are well structured, and they are also very short, besides the fact that one is working five days a week, sometimes there is not much time left, but in this case, they are very short and precise, and they could be solved in the medium term. » (Jorge)*	*« They gave us the confidence to ask questions whenever we had doubts. […] Unlike other training programs, we always received support. That is why we felt we learned more: we were encouraged to keep asking questions, and they supported us throughout the implementation process.» (Jesús)*
T3: Both	*«The teacher was […] very innovative, very creative. I always liked the fact that she had a lot of didactic and technological resources, so that really appealed to her. As a teacher, it attracted a lot of attention, because you know that to create these kinds of classes you invest a lot of time. » (Ruth)*	*«They were quite good, because they served as a resource for those contents that were a little unclear to us in the class […]. We would watch them again, review the content and use them to fill in the portfolio guides that the teacher gave us. » (Ruth)*	*«I consider them necessary, […] because we also needed to consolidate the knowledge through the guides that we had. That we had to present the portfolio and the exercises, I consider that it was enough, there was not much overload […]. » (Ana)*	*« This project really provided ongoing support for teachers. […] You can receive training, but questions always arise about how to implement […]. Throughout the program, I felt a great deal of support from the facilitators and from the teaching resources they provided. That was the main difference.» (Ruth)*

Figure A12: Questions on program activities



Word	Frequency
excellent	12
liked	11
fine	10
good	10
like	3
nice	3
seemed	3

Figure A13: Word cloud of improvement-related feedback and co-occurrences associated with the term "everything."

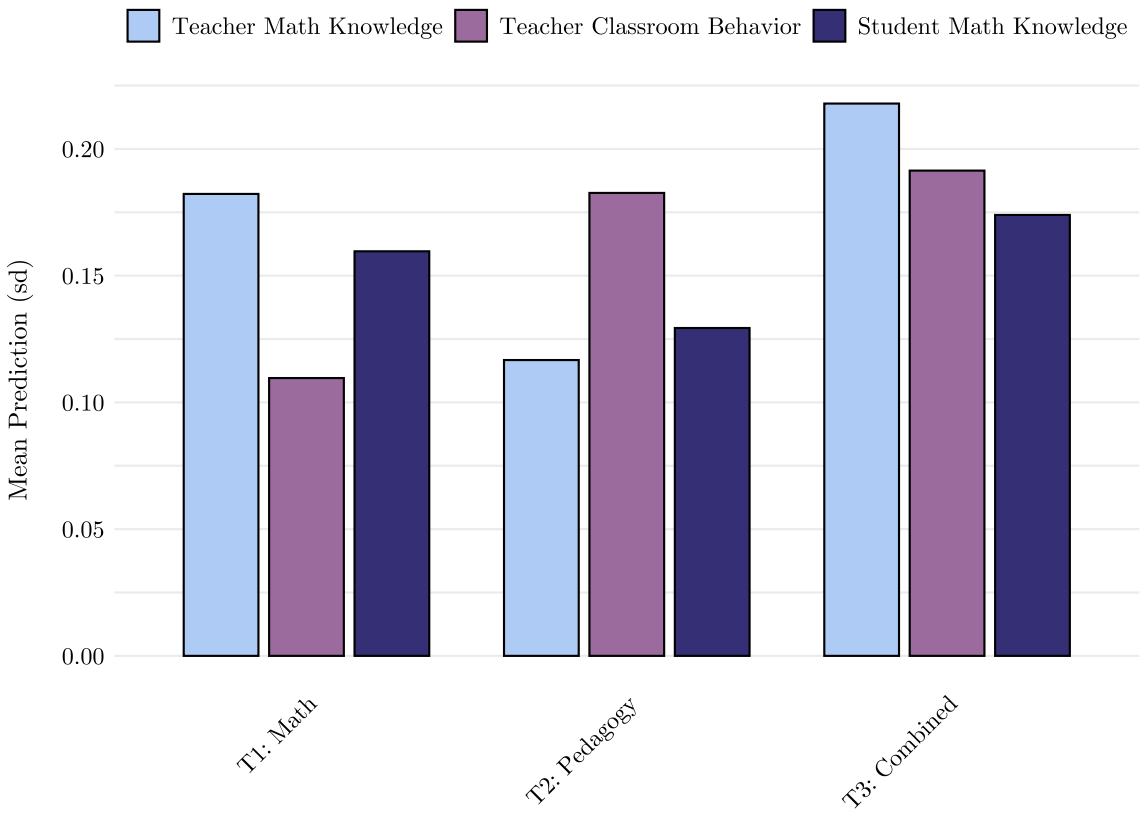


Figure A14: Predictions on program impact

Illustration based on a prediction survey with 38 education researchers and practitioners. The figure shows mean predictions of effect sizes in standard deviations per treatment. For each treatment, respondents predicted effects on teacher content knowledge, teacher classroom behavior and student content knowledge.